

Definition: Let S be the sample space associated with a given random experiment. Let $X = X(s)$ and $Y = Y(s)$ be two functions each assigning a real no to each outcome $s \in S$. Then we call the pair (X, Y) a two dimensional random variable.

Ex: 1 Height and weight of a randomly chosen person, Age and blood pressure, Crop yield and rain fall, IQ and nutrition of an individual.

Discrete Random Variables:

Joint probability distribution or Joint probability mass function

If X and Y are two discrete random variables, we define the joint probability function of X and Y by

$$P(X=x, Y=y) = f(x, y)$$

where $f(x, y)$ satisfy the conditions.

$$f(x, y) \geq 0 \text{ and } \sum_x \sum_y f(x, y) = 1$$

The second condition means that the sum ~~of~~ over all the values of x and y is equal to one.

Suppose $X = \{x_1, x_2, \dots, x_m\}$ and $Y = \{y_1, y_2, \dots, y_n\}$

then $P(X=x_i, Y=y_j) = f(x_i, y_j)$ denoted by p_{ij}

The set of triplets $(x_i, y_j, p(x_i, y_j))$, $i, j = 1, 2, \dots, (i=m), (j=n)$ is called the joint probability distribution of X and Y .

These values are presented in the form of a two way table called the joint probability table.

$x \backslash y$	y_1	y_2	$\dots$	y_n	$\text{Sum}_{P(x=x_i)} f(x_i)$
x_1	p_{11}	p_{12}	$\dots$	p_{1n}	$f(x_1)$
x_2	p_{21}	p_{22}	$\dots$	p_{2n}	$f(x_2)$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
x_m	p_{m1}	p_{m2}	$\dots$	p_{mn}	$f(x_m)$
$\text{Sum}_{P(y=y_j)} g(y_j)$	$g(y_1)$	$g(y_2)$	$\dots$	$g(y_n)$	1

Marginal probability distributions:

In the joint probability table, $f(x_1), f(x_2), \dots, f(x_m)$ respectively represents the sum of all ^{the} entries in the first row, second row $\dots$ m^{th} row. $g(y_1), g(y_2), \dots, g(y_n)$ respectively represents the sum of all the entries in the first column, second column $\dots$ n^{th} column.

That is, $f(x_1) = p_{11} + p_{12} + \dots + p_{1n}$; $g(y_1) = p_{11} + p_{21} + \dots + p_{m1}$

$$f(x_2) = p_{21} + p_{22} + \dots + p_{2n}; \quad g(y_2) = p_{12} + p_{22} + \dots + p_{m2}$$

$$f(x_m) = p_{m1} + p_{m2} + \dots + p_{mn}; \quad g(y_n) = p_{1n} + p_{2n} + \dots + p_{mn}$$

$\{f(x_1), f(x_2), \dots, f(x_m)\}$ and $\{g(y_1), g(y_2), \dots, g(y_n)\}$ are called marginal probability distributions (marginal probability mass functions) of X and Y respectively.

(Since these values are from the margins of the table the word "marginal" is being used).

It should be noted that $f(x_1) + f(x_2) + \dots + f(x_m) = 1$ and $g(y_1) + g(y_2) + \dots + g(y_n) = 1$

This is equivalent to writing

$$\sum_{i=1}^m \sum_{j=1}^n f(x_i, y_j) = \sum_{i=1}^m \sum_{j=1}^n p_{ij} = 1$$

It seems that the total of all entries in the joint probability table is equal to 1.

Conditional probability Distributions

Defn:- Given discrete random variables X and Y with joint probability mass function $f(x, y)$, the conditional probability mass function of Y given $X=x$ is

$$f_{Y|X}(y) = f_{XY}(x, y) / f_X(x), \text{ for } f_X(x) > 0.$$

Independence random variables

The discrete random variables X and Y are said to be independent random variables if

$$P(X=x, Y=y) = P(X=x) \cdot P(Y=y) \text{ and conversely.}$$

$$\text{That is } P(X=x_i, Y=y_j) = P(X=x_i) \cdot P(Y=y_j)$$

This is equivalent to $f(x_i) g(y_j) = P_{ij}$ in the joint probability table.

That is to say that X and Y are independent if each entry P_{ij} in the table is equal to the product of its marginal entries otherwise X and Y are said to be dependent.

Expectation and Variance

If X and Y are two discrete random variables having the joint probability function $f(x, y)$, then the expectations of X and Y are defined as follows.

$$M_X = E(X) = \sum_x \sum_y x f(x, y) = \sum_i x_i f(x_i)$$

$$M_Y = E(Y) = \sum_x \sum_y y f(x, y) = \sum_j y_j g(y_j)$$

$$\text{Further } E(XY) = \sum_{i,j} x_i y_j P_{ij}$$

Problems: ① The joint distribution of two random variables X and Y as follows.

$X \backslash Y$	-4	2	7
1	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$
5	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$

① $E(X) = \sum x_i \cdot f(x_i) = 1 \times \frac{1}{2} + 5 \times \frac{1}{2} = 3$
 $E(Y) = \sum y_j \cdot f(y_j) = -4 \times \frac{3}{8} + 2 \times \frac{3}{8} + 7 \times \frac{1}{4} = 1$
 Thus $\mu_X = E(X) = 3$ & $\mu_Y = E(Y) = 1$
 ② $E(XY) = \sum x_i y_j \cdot f(x_i, y_j) = \mu_{XY}$
 $= 1 \times -4 \times \frac{1}{8} + 1 \times 2 \times \frac{1}{4} + 1 \times 7 \times \frac{1}{8} + 5 \times -4 \times \frac{1}{4} + 5 \times 2 \times \frac{1}{8} + 5 \times 7 \times \frac{1}{8} = \frac{3}{2}$

③ $\sigma_X^2 = E(X^2) - \mu_X^2$ and $\sigma_Y^2 = E(Y^2) - \mu_Y^2$
 $E(X^2) = \sum x_i^2 \cdot f(x_i) = 1 \times \frac{1}{2} + 25 \times \frac{1}{2} = 13$
 $E(Y^2) = \sum y_j^2 \cdot f(y_j) = 16 \times \frac{3}{8} + 4 \times \frac{3}{8} + 49 \times \frac{1}{4} = 13$
 $\therefore \sigma_X^2 = 13 - 3^2 = 4$, $\sigma_Y^2 = 13 - 1^2 = 12$
 $\therefore \sigma_X = 2$, $\sigma_Y = 2\sqrt{3}$

Determine the marginal probability distributions of X and Y .

Also, verify that X and Y are independent random variables. Determine the following ② $E(X)$ and $E(Y)$ ③ $E(XY)$ ④ σ_X and σ_Y

Sol:- The distributions (marginal distributions) of X and Y is as follows. The distribution is obtained by adding all the respective row entries and also the respective column entries.

Distribution of X

x_i	1	5
$f(x_i)$	$\frac{1}{2}$	$\frac{1}{2}$

Distribution of Y

y_j	-4	2	7
$g(y_j)$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{4}$

If X and Y are independent random variables, we must have

$$f(x_i) \cdot g(y_j) = p_{ij}$$

It can be seen that $f(x_1)g(y_1) = \frac{1}{2} \times \frac{3}{8} = \frac{3}{16}$ and $p_{11} = \frac{1}{8}$

i.e., $f(x_1)g(y_1) \neq p_{11}$; $f(x_2)g(y_1) \neq p_{21}$

Similarly $f(x_1)g(y_2) \neq p_{12}$; $f(x_2)g(y_2) \neq p_{22}$

$f(x_1)g(y_3) \neq p_{13}$, $f(x_2)g(y_3) \neq p_{23}$

Hence we conclude that X and Y are dependent random variables.

② The joint probability distribution table for two random variables X and Y is as follows:

$X \backslash Y$	-2	-1	4	5
1	0.1	0.2	0	0.3

Mean and Variance from joint distribution

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If the marginal probability distribution of X has the probability mass function $f_X(x)$, then

$$E(X) = \mu_X = \sum_x x f_X(x) = \sum_x x \sum_{y \in R_x} f_{XY}(x, y) = \sum_x \sum_{y \in R_x} x f_{XY}(x, y) = \sum_x \sum_{y \in R_x} (x - \mu_X) f_{XY}(x, y)$$

and $V(X) = \sigma_X^2 = \sum_x (x - \mu_X)^2 f_X(x) = \sum_x (x - \mu_X)^2 \sum_{y \in R_x} f_{XY}(x, y)$
 R_x denotes the set of all the points in the range of (X, Y) for which $X=x$.
 where R denotes the set of all points in the range of (X, Y) .

Defn: Let R_x denote the set of all points in the range of (X, Y) for which $X=x$. The conditional mean of Y given $X=x$, denoted as $E(Y/x)$ or $\mu_{Y/x}$, is

$$E(Y/x) = \sum_{y \in R_x} y f_{Y/x}(y)$$

and the conditional variance of Y given $X=x$, denoted as $V(Y/x)$ or $\sigma_{Y/x}^2$, is

$$V(Y/x) = \sum_{y \in R_x} (y - \mu_{Y/x})^2 f_{Y/x}(y) = \sum_{y \in R_x} y^2 f_{Y/x}(y) - \mu_{Y/x}^2$$

Further $E(XY) = \sum_{i=1}^m \sum_{j=1}^n x_i y_j p_{ij}$

Miscellaneous problem

(P) A fair coin is tossed thrice. The random variables X and Y are defined follows, $X=0$ or 1 according as head or tail occurs on the first toss.

Y = Number of heads

- (a) Determine the distributions of X and Y (b) Determine the joint distribution of X and Y (c) Obtain the expectations of X, Y and XY . Also find S.D.s of X and Y .

Sol: The sample space S and the association of random variables X and Y is given by the following table:

S :	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
X :	0	0	0	0	1	1	1	1
Y :	3	2	2	1	2	1	1	0

$$P(X=0) = \frac{4}{8} = \frac{1}{2}; P(X=1) = \frac{4}{8} = \frac{1}{2}$$

$$P(Y=0) = \frac{1}{8}; P(Y=1) = \frac{3}{8}$$

$$P(Y=2) = \frac{3}{8}; P(Y=3) = \frac{1}{8}$$

(a) The probability distribution of X and Y is found as follows

Thus we have the following probability distributions of X and Y

x_i	0	1
$f(x_i)$	$\frac{1}{2}$	$\frac{1}{2}$

y_j	0	1	2	3
$g(y_j)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

⑤ The joint distribution of X and Y is found by computing

$P_{ij} = P(X=x_i, Y=y_j)$ where we have $x_1=0, x_2=1$ and $y_1=0, y_2=1, y_3=2, y_4=3$

$P_{11} = P(X=0, Y=0) = 0$ ($X=0$ implies that there is a head turn out and Y the total number of heads 0 is impossible)

$P_{12} = P(X=0, Y=1) = \frac{1}{8}$ (corresponding to the outcome HTT)

$P_{13} = P(X=0, Y=2) = \frac{1}{4}$ (outcomes are HHT and HTH)

$P_{14} = P(X=0, Y=3) = \frac{1}{8}$ (outcome is HHH)

$P_{21} = P(X=1, Y=0) = \frac{1}{8}$ (outcome is TTT)

$P_{22} = P(X=1, Y=1) = \frac{1}{4}$ (outcomes are THT, TTH)

$P_{23} = P(X=1, Y=2) = \frac{1}{8}$ (outcome is TTH)

$P_{24} = P(X=1, Y=3) = 0$ (since the outcome is impossible)

The required joint probability distribution of X and Y is as follows:

$X \backslash Y$	0	1	2	3	$f(x_i)$
0	0	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{2}$
1	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$	0	$\frac{1}{2}$
$g(y_j)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	1

⑥ $\mu_x = E(X) = \sum x f_x(x) = 0 \times \frac{1}{2} + 1 \times \frac{1}{2} = \frac{1}{2}$

$\mu_y = E(Y) = \sum y f_y(y) = 0 \times \frac{1}{8} + 1 \times \frac{3}{8} + 2 \times \frac{3}{8} + 3 \times \frac{1}{8} = \frac{3}{2}$

$\mu_{xy} = E(XY) = \sum_{x,y} xy p_{ij} = 0 + (0 + \frac{1}{4} + \frac{2}{8} + 0) = \frac{1}{2}$

$\sigma_x^2 = V(X) = E(X^2) - \mu_x^2$ and $\sigma_y^2 = V(Y) = E(Y^2) - \mu_y^2$

$\sigma_x^2 = (0 + \frac{1}{2}) - \frac{1}{4} = \frac{1}{4}$

$\sigma_y^2 = (0 + \frac{3}{8} + \frac{3}{2} + \frac{9}{8}) - \frac{9}{4} = 3 - \frac{9}{4} = \frac{3}{4}$

$\therefore \sigma_x = \sqrt{\frac{1}{4}} = \frac{1}{2}$ and $\sigma_y = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$

Q. Determine the marginal probability distributions of X and Y .

Q. Verify that X and Y are dependent random variables

Q. Find $P(X+Y > 0)$ Q. Expectations of X, Y and XY Q. S.D's of X, Y

Ans: (A) $E(X)=1.4, E(Y)=1.4, E(XY)=0.9$ (B) $\sigma_x=0.49, \sigma_y=3$

Sol.: Marginal distributions of X and Y are got by adding all respective row entries and the respective column entries.

x_i	1	2
$f(x_i)$	0.6	0.4

y_j	-2	-1	4	5
$g(y_j)$	0.3	0.3	0.1	0.3

If X and Y are independent random variables, we must have

$$f(x_i) g(y_j) = p_{ij}$$

It can be seen that $f(x_1)g(y_1) = 0.6 \times 0.3 = 0.18$ and $p_{11} = 0.1$

$$\text{ie } f(x_1)g(y_1) \neq p_{11}$$

Similarly for others also the condition is not satisfied.

Hence we conclude that X and Y are dependent random variables

We have $X = \{x_i\} = \{x_1, x_2\} = \{1, 2\}$ respectively

$Y = \{y_j\} = \{y_1, y_2, y_3, y_4\} = \{-2, -1, 4, 5\}$ respectively

Also $p_{11} = 0.1$; $p_{12} = 0.2$, $p_{13} = 0$, $p_{14} = 0.3$, $p_{21} = 0.2$, $p_{22} = 0$

$p_{23} = 0.1$, $p_{24} = 0$

$X+Y > 0$ is possible when (X, Y) take the values

$(x_1, y_3) = (1, 4)$; $(x_1, y_4) = (1, 5)$; $(x_2, y_2) = (2, -1)$; $(x_2, y_3) = (2, 4)$
and $(x_2, y_4) = (2, 5)$

Hence $P(X+Y > 0) = p_{13} + p_{14} + p_{22} + p_{23} + p_{24} = 0.5$

(3) Suppose X and Y are independent random variables with the following respective distribution, find the joint distribution of

X and Y :

x_i	1	2
$f(x_i)$	0.7	0.3

y_j	-2	5	8
$h(y_j)$	0.3	0.5	0.2

Sol:- Since X and Y are independent, the joint probability distribution

$f(x, y)$ is obtained by using the definition $f(x_i)g(y_j) = P_{ij}$

P_{ij} are obtained on multiplication of the marginal entries.

The joint probability distribution of X and Y is as follows

$X \backslash Y$	$y_1 = -2$	$y_2 = 5$	$y_3 = 8$	$f(x_i)$
$x_1 = 1$	P_{11}	P_{12}	P_{13}	0.7
$x_2 = 2$	P_{21}	P_{22}	P_{23}	0.3
$g(y_j)$	0.3	0.5	0.2	1

$$P_{11} = f(x_1)g(y_1) = 0.7 \times 0.3 = 0.21, P_{12} = f(x_1)g(y_2) = 0.7 \times 0.5 = 0.35$$

$$P_{13} = f(x_1)g(y_3) = 0.7 \times 0.2 = 0.14, P_{21} = 0.3 \times 0.3 = 0.09$$

$$P_{22} = 0.3 \times 0.5 = 0.15; P_{23} = 0.3 \times 0.2 = 0.06$$

$X \backslash Y$	-2	5	8	$f(x_i)$
1	0.21	0.35	0.14	0.7
2	0.09	0.15	0.06	0.3
$g(y_j)$	0.3	0.5	0.2	1

④ The joint probability distribution of two discrete random variables X and Y is given by $f(x, y) = k(2x + y)$ where x and y are integers such that $0 \leq x \leq 2, 0 \leq y \leq 3$.

⑥ Find the value of the constant k ⑦ Find the marginal probability distributions of X and Y . ⑧ Show that the random variables X and Y are independent. ⑨ Compute the following $E(X), E(Y), E(XY), E(X^2), E(Y^2), \sigma_X, \sigma_Y$.

Sol:- $X = \{x_i\} = \{0, 1, 2\}$ and $Y = \{y_j\} = \{0, 1, 2, 3\}$

$f(x, y) = k(2x + y)$ and the joint probability distribution table is formed as follows.

$X \backslash Y$	0	1	2	3	Sum
0	0	k	$2k$	$3k$	$6k$
1	$2k$	$3k$	$4k$	$5k$	$14k$
2	$4k$	$5k$	$6k$	$7k$	$22k$

⑩ We must have $42k = 1$

$$\therefore k = \frac{1}{42}$$

③ Marginal probability distributions are as follows.

x_i	0	1	2
$f(x_i)$	$6/42$ $= 1/7$	$14/42$ $= 1/3$	$22/42$ $= 11/21$

y_j	0	1	2	3
$g(y_j)$	$6/42$ $= 1/7$	$9/42$ $= 3/14$	$12/42$ $= 2/7$	$15/42$ $= 5/14$

④ It can be easily seen that $f(x_i)g(y_j) \neq p_{ij}$

① $P(X=1, Y=2)$ ② $P(X=2, Y=1)$ ③ $P(X \geq 1, Y \leq 2)$ ④ $P(X+Y > 2)$

Sol: ① $P(X=1, Y=2) = f(x_2, y_3) = p_{23} = 4/42 = 2/21$

② $P(X=2, Y=1) = f(x_3, y_2) = p_{32} = 5/42$

③ $P(X \geq 1, Y \leq 2)$

(X, Y) take the values $(1,0), (1,1), (1,2), (2,0), (2,1), (2,2)$

They are respectively $(x_2, y_1), (x_2, y_2), (x_2, y_3), (x_3, y_1), (x_3, y_2), (x_3, y_3)$

$\therefore P(X \geq 1, Y \leq 2) = p_{21} + p_{22} + p_{23} + p_{31} + p_{32} + p_{33} = 4/7$

④ To find $P(X+Y > 2)$, (X, Y) take the values

$(0,3), (1,2), (1,3), (2,1), (2,2), (2,3)$

They are respectively $(x_1, y_4), (x_2, y_3), (x_2, y_4), (x_3, y_2), (x_3, y_3), (x_3, y_4)$

$\therefore P(X+Y > 2) = p_{14} + p_{23} + p_{24} + p_{32} + p_{33} + p_{34} = 5/7$

Exercises for Section 5.1:

⑤.1 Show that the following function satisfies the properties of a joint probability mass function.

x	y	$f_{xy}(x,y)$
1	1	$1/4$
1.5	2	$1/8$
1.5	3	$1/4$
2.5	4	$1/4$
3	5	$1/8$

(i) Determine the following probabilities

① $P(X < 2.5, Y < 3)$ ② $P(X < 2.5)$

③ $P(Y < 3)$ ④ $P(X > 1.8, Y > 4.7)$

(ii) Determine the marginal probability distribution of the random variable $X+Y$

(iii) Determine the conditional probability

distribution of y given that $X=1.5$ ⑤ Determine the conditional

probability distribution of X given that $Y=2$ ⑥ Are X and Y

independent? (vi) Determine $E(X)$, $E(Y)$ and $E(Y|X=1.5)$.

Sol: $P(x_i, y_j) \geq 0 \quad \forall (x_i, y_j)$

$X \backslash Y$	1	2	3	4	5	$P(X=x_i)$
1	$\frac{1}{4}$	0	0	0	0	$\frac{1}{4}$
1.5	0	$\frac{1}{8}$	$\frac{1}{4}$	0	0	$\frac{3}{8}$
2.5	0	0	0	$\frac{1}{4}$	0	$\frac{1}{4}$
3	0	0	0	0	$\frac{1}{8}$	$\frac{1}{8}$
$P(Y=y_j)$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{8}$	1

Observe that:

$$\sum_{j=1}^5 \sum_{i=1}^4 f(x_i, y_j) = 1$$

and $f(x_i, y_j) \geq 0 \quad \forall (x_i, y_j)$

$\therefore$ the given function is a joint probability distribution function of X and Y .

① $P(X < 2.5, Y < 3) = P(X=1, Y=1) + P(X=1.5, Y=1) + P(X=1, Y=2)$
 $+ P(X=1.5, Y=2) = \frac{1}{4} + 0 + 0 + \frac{1}{8} = \frac{3}{8}$

② $P(X < 2.5) = P(X=1, Y=1) + P(X=1, Y=2) + P(X=1, Y=3) + P(X=1, Y=5)$
 $+ P(X=1.5, Y=1) + P(X=1.5, Y=2) + P(X=1.5, Y=3)$
 $+ P(X=1.5, Y=4) + P(X=1.5, Y=5)$
 $= \frac{1}{2} + 0 + 0 + 0 + 0 + 0 + \frac{1}{8} + \frac{1}{4} + 0 + 0 = \frac{5}{8}$

③ $P(Y < 3) = P(X=1, Y=1) + P(X=1, Y=2) + P(X=1.5, Y=1)$
 $+ P(X=1.5, Y=2) + P(X=2.5, Y=1) + P(X=2.5, Y=2)$
 $+ P(X=3, Y=1) + P(X=3, Y=2) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$

④ $P(X > 1.8, Y > 4.7) = P(X=2.5, Y=5) + P(X=3, Y=5) = \frac{1}{8}$

⑤

Distribution of X

x_i	1	1.5	2.5	3
$P(x_i)$	$\frac{1}{4}$	$\frac{3}{8}$	$\frac{1}{4}$	$\frac{1}{8}$

Distribution of Y

y_j	1	2	3	4	5
$P(y_j)$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{8}$

(iii) The conditional probability distribution of y given $x = x_i$ is

$$f_{y|x}(y) = f_{xy}(x, y) / f_x(x)$$

$$(vi) E(x) = \mu_x = \sum x f_x(x)$$

$$= 1 \times \frac{1}{4} + \frac{3}{2} \times \frac{3}{8} + \frac{5}{2} \times \frac{1}{4} + \frac{1}{4} + \frac{9}{16} + \frac{5}{8} + \frac{3}{8} = \frac{4+9+11}{16} = \frac{24}{16} = 1.5$$

$$f_{y|1.5}(y_1) = \frac{P(1.5, 1)}{P(1.5)} = \frac{0}{3/8} = 0$$

$$f_{y|1.5}(y_2) = \frac{P(1.5, 2)}{P(1.5)} = \frac{\frac{1}{8}}{3/8} = \frac{1}{3}$$

$$f_{y|1.5}(y_3) = \frac{P(1.5, 3)}{P(1.5)} = \frac{\frac{1}{4}}{3/8} = \frac{2}{3}$$

$$f_{y|1.5}(y_4) = \frac{P(1.5, 4)}{P(1.5)} = \frac{0}{3/8} = 0$$

$$f_{y|1.5}(y_5) = \frac{P(1.5, 5)}{P(1.5)} = \frac{0}{3/8} = 0$$

$$E(y) = \mu_y = \sum y f_y(y) = \frac{1}{4} + \frac{2}{8} + \frac{3}{4} + 1 + \frac{5}{8} = \frac{2+2+6+8+5}{8} = \frac{23}{8} = 2.875$$

$$E(y|x=1.5) = \sum_{k=1}^5 y f_{y|x}(y) = \sum_{k=1}^5 y \frac{f_{xy}(x, y)}{f_x(1.5)} = \sum_{k=1}^5 y \frac{f_{xy}(1.5, y)}{f_x(1.5)} = \frac{2 \times \frac{1}{8} + 3 \times \frac{1}{4} + 4 \times \frac{1}{8}}{\frac{3}{8}} = \frac{2+6+4}{3} = \frac{12}{3} = 4$$

y	1	2	3	4	5
$f_{y 1.5}(y_i)$	0	$\frac{1}{3}$	$\frac{2}{3}$	0	0

x	1	1.5	2.5	3	4
$f_{x 2}(x_i)$	$\frac{0}{1/8} = 0$	$\frac{1/8}{1/8} = 1$	$\frac{0}{1/8} = 0$	$\frac{0}{1/8} = 0$	

(v) If x and y are independent random variables, we must have

$$P_{ij} = f(x_i) \times g(y_j)$$

$$P(x=1, y=1) = \frac{1}{4} \neq P(x=1) \times P(y=1) = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$

$\therefore x$ and y are not independent.

(2) Determine the value of C that makes the function $f(x, y) = C(x+y)$ a joint probability mass function over the nine points with $x=1, 2, 3$ and $y=1, 2, 3$.

(3) Determine the following probabilities:

(i) $P(x=1, y < 4)$ (ii) $P(x=1)$ (iii) $P(y=2)$ (iv) $P(x < 2, y < 2)$.

(b) The marginal distribution of the random variable x

- (a) The conditional probability distribution of Y given that $X=1$
 (b) The conditional probability distribution of X given that $Y=2$
 (c) Are X and Y independent. (i) Determine $E(X)$, $E(Y)$, $V(X)$ and $V(Y)$
 (ii) Also determine $E(Y/X=1)$.

Sol: 1-

$X \backslash Y$	1	2	3	$f_{X(x)}$
1	$2c$	$3c$	$4c$	$9c$
2	$8c$	$4c$	$5c$	$12c$
3	$4c$	$5c$	$6c$	$15c$
$g(y)$	$9c$	$12c$	$15c$	1

We must have,
 $9c + 12c + 15c = 1 \Rightarrow c = \frac{1}{36}$

$X \backslash Y$	1	2	3	
1	$\frac{1}{18}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{9}{36} = \frac{1}{4}$
2	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{5}{36}$	$\frac{12}{36} = \frac{1}{3}$
3	$\frac{1}{9}$	$\frac{5}{36}$	$\frac{1}{6}$	$\frac{15}{36} = \frac{5}{12}$
	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{5}{12}$	1

- (a) (i) $P(X=1, Y < 4) = P(X=1, Y=1) + P(X=1, Y=2) + P(X=1, Y=3)$
 $= \frac{1}{18} + \frac{1}{12} + \frac{1}{9} = \frac{1}{4} = 0.25$
 (ii) $P(X=1) = \frac{1}{4} = 0.25$
 (iii) $P(Y=2) = P(X=1, Y=2) + P(X=2, Y=2) + P(X=3, Y=2)$
 $= \frac{1}{12} + \frac{1}{9} + \frac{5}{36} = \frac{1}{3} = 0.3333$
 (iv) $P(X < 2, Y < 2) = P(X=1, Y=1) = \frac{1}{18} = 0.0556$

(b) Marginal probability distribution of X .

X	1	2	3
$P(X=x_i)$	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{5}{12}$

Marginal probability distribution of Y

Y	1	2	3
$P(Y=y_i)$	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{5}{12}$

(c) $f_{Y/X}(y_j) = \frac{f_{XY}(x_i, y_j)}{f_X(x_i)}, f_X(x_i) > 0$
 $f_{Y/1}(y_1) = \frac{f_{XY}(1,1)}{f_X(1)} = \frac{1/18}{1/4} = \frac{2}{9} = 0.2222$
 $f_{Y/1}(y_2) = \frac{f_{XY}(1,2)}{f_X(1)} = \frac{1/12}{1/4} = \frac{1}{3} = 0.3333$
 $f_{Y/1}(y_3) = \frac{f_{XY}(1,3)}{f_X(1)} = \frac{1/9}{1/4} = \frac{4}{9} = 0.4444$

Conditional probability of Y given $X=1$ is

Y	1	2	3
$P(Y)$	0.2223	0.3333	0.4444

(d) $E(X) = \sum x f_X(x) = \frac{1}{4} + \frac{2}{3} + \frac{15}{12} = \frac{26}{12} = 2.167$
 $E(Y) = \sum y f_Y(y) = \frac{1}{4} + \frac{2}{3} + \frac{15}{12} = 2.167$
 $V(X) = \sum (x - \mu)^2 f_X(x) = (1-2.167)^2 \times \frac{1}{4} + (2-2.167)^2 \times \frac{1}{3} + (3-2.167)^2 \times \frac{5}{12} = 0.63$
 $V(Y) = (1-2.167)^2 \times \frac{1}{4} + (2-2.167)^2 \times \frac{1}{3} + (3-2.167)^2 \times \frac{5}{12} = 0.639$

② Conditional probability distribution of X given $Y=2$ is

$$f_{X/2}(x_i) = \frac{f_{XY}(x_i, 2)}{f_Y(2)} = \frac{f_{XY}(x_i, 2)}{f_Y(2)}, \quad i=1, 2, 3$$

$$f_{X/2}(x_1) = \frac{f_{XY}(1, 2)}{f_Y(2)} = \frac{1/12}{1/3} = \frac{1}{12} \times \frac{3}{1} = \frac{1}{4} = 0.25$$

$$f_{X/2}(x_2) = \frac{f_{XY}(2, 2)}{f_Y(2)} = \frac{1/9}{1/3} = \frac{1}{9} \times \frac{3}{1} = \frac{1}{3} = 0.3333$$

$$f_{X/2}(x_3) = \frac{f_{XY}(3, 2)}{f_Y(2)} = \frac{5/36}{1/3} = \frac{5}{36} \times \frac{3}{1} = \frac{5}{12} = 0.4167$$

∴ Conditional probability distribution in the form of a table is

X	1	2	3
$f_{X/2}(x_i)$	0.25	0.3333	0.4167

$$\begin{aligned} \textcircled{3} E(Y|X=1) &= M_{Y|1} = \sum_{j=1}^3 y_j f_{Y|1}(y_j) \\ &= \sum_{j=1}^3 y_j \frac{f_{XY}(X=1, y_j)}{f_X(1)} \\ &= \frac{1 \cdot \frac{1}{12} + 2 \cdot \frac{1}{12} + 3 \cdot \frac{1}{12}}{\frac{1}{3}} \\ &= \frac{\frac{1}{12} + \frac{2}{12} + \frac{3}{12}}{\frac{1}{3}} = \frac{\frac{6}{12}}{\frac{1}{3}} = \frac{1}{2} \times 3 = 1.5 \end{aligned}$$

③ $P(X=x_i, Y=y_j) = P(X=x_i) \cdot P(Y=y_j)$, if X and Y are independent

$$P(X=1, Y=1) = 0.0556 \neq 0.25 \times 0.25 = 0.0625$$

∴ X and Y are not independent.

(S.13) Four electronic printers are selected from a large lot of damaged printers. Each printer is inspected and classified as containing either a major or a minor defect. Let the random variables X and Y denote the no. of printers with major and minor defects, respectively. Determine the range of the joint probability distribution of X and Y .

Sol: $X \rightarrow$ no. of printers with major defects
 $Y \rightarrow$ no. of printers with minor defects

$$X = \{0, 1, 2, 3, 4\}, \quad Y = \{0, 1, 2, 3, 4\}$$

$$X \geq 0, \quad Y \geq 0, \quad X + Y \leq 4$$

- 5.14 In the transmission of digital information, the probability that a bit has high, moderate, and low distortion is 0.01, 0.10 and 0.95, respectively. Suppose that the three bits are transmitted and that the amount of distortion of each bit is assumed to be independent. Let x denote the number of bits with high and moderate distortion out of the three, respectively. Determine (a) $f_{xy}(x, y)$ (b) $f_x(x)$, (c) $f_{y|1}(y)$ (d) Are x and y independent?

Sol: $h \rightarrow$ high distortion, $m \rightarrow$ moderate distortion, $l \rightarrow$ low distortion

$$P(h) = 0.01, P(m) = 0.10, P(l) = 0.95$$

$X \rightarrow$ no of bits with high distortion,

$Y \rightarrow$ no of bits with moderate distortion

$$X \geq 0, Y \geq 0, X + Y \leq 3$$

$f_{xy}(x, y)$ (joint probability distribution of x and y)

$X \backslash Y$	0	1	2	3	$P(X=x_i)$
0	lll 0.8574	mlh 0.0025	mmh 0.0005	mmm 0.0001	0.9582
1	hll 0.009	hml 0.0009	hmm 0.0001	0	0.01005
2	hhl 0.000095	hhl 0.00001	0	0	0.0001
3	hhh 0.000001	0	0	0	0.000001
$P(Y=y_j)$	0.866496	0.09126	0.0096	0.001	0.968352

$f_x(x)$ (marginal distribution of x)

$x:$

$$f_x(x): 0.9582 \quad 0.01005 \quad 0.0001 \quad 0.000001$$

$f_{y|1}(y)$ (Conditional probability distribution of Y given that $X=1$)

$$f_{y|1}(y_1) = \frac{f_{xy}(1, 0)}{f_x(1)} = \frac{0.009}{0.01005} = 0.89522, \quad f_{y|1}(y_2) = \frac{f_{xy}(1, 1)}{f_x(1)} = \frac{0.0009}{0.01005} = 0.0945$$

$$f_{y|1}(y_3) = \frac{f_{xy}(1, 2)}{f_x(1)} = \frac{0.0001}{0.01005} = 0.00995, \quad f_{y|1}(y_4) = \frac{f_{xy}(1, 3)}{f_x(1)} = 0$$

Are x and y independent?

We must have $P(X=1, Y=1) = P(X=1) \cdot P(Y=1) \Rightarrow 0.00095 \neq 0.09126 \times 0.01005 (0.000917)$

problems

Q1. Let X and Y be independent Random Variables, X takes values 2, 5, 7 with probability $\frac{1}{2}, \frac{1}{4}$ & $\frac{1}{4}$ respectively and Y takes values 3, 4, 5 with probability $\frac{1}{3}$ each. Find the joint distribution of X and its correlation coefficient and also find the distribution of

$$Z = X + Y$$

Sol:

$Y \backslash X$	3	4	5	$f_{Y}(y)$
2	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{2}$
5	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{4}$
7	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{4}$
$g_Y(y)$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	1

$$\text{COV}(X, Y) = E(XY) - E(X)E(Y)$$

$$= 16 - 4 \times 4 = 0$$

$$\rho(X, Y) = \frac{\text{COV}(X, Y)}{\sigma_X \sigma_Y}$$

$$E(X) = 2 \times \frac{1}{2} + 5 \times \frac{1}{4} + 7 \times \frac{1}{4} = 4$$

$$E(Y) = 3 \times \frac{1}{3} + 4 \times \frac{1}{3} + 5 \times \frac{1}{3} = 4$$

$$E(XY) = 2 \times 3 \times \frac{1}{6} + 2 \times 4 \times \frac{1}{6} + 2 \times 5 \times \frac{1}{6} + 5 \times 3 \times \frac{1}{12} + 5 \times 4 \times \frac{1}{12} + 5 \times 5 \times \frac{1}{12} + 7 \times 3 \times \frac{1}{12} + 7 \times 4 \times \frac{1}{12} + 7 \times 5 \times \frac{1}{12}$$

$$= 1 + \frac{4}{3} + \frac{5}{3} + \frac{5}{3} + \frac{5}{3} + \frac{25}{12} + \frac{7}{4} + \frac{7}{3} + \frac{35}{12}$$

$$= 12 + 16 + 20 + 15 + 20 + 25 + 21 + 28 + \dots$$

$$= \frac{12 \times 9 \times 46}{12 \times 9 \times 46} = \frac{48}{3} = 16$$

$$= \frac{192}{12} = \frac{16}{1} = 16$$

Q2. Two refills for a pen are selected at random from a box that contains 3 blue, 2 red and 3 green refills. If X is the no. of blue refills and Y the no. of red refills, find the JPMP of X and Y & probability of $X + Y \leq 1$ and the covariance of X & Y .

Sol 1. - $P(X=0, Y=0) = GG = \frac{3}{8} \times \frac{2}{7} = \frac{3}{28}$

$$P(X=0, Y=1) = RG, GR = \frac{2}{8} \times \frac{3}{7} \times 2$$

$$P(X=0, Y=2) = RR = \frac{2}{8} \times \frac{1}{7} = \frac{1}{28}$$

$$P(X=1, Y=0) = BG, GB = \frac{3}{8} \times \frac{2}{7} \times 2 = \frac{6}{28}$$

$$P(X=1, Y=1) = BR, RB = \frac{3}{8} \times \frac{2}{7} \times 2 = \frac{6}{28}$$

$$P(X=1, Y=2) = \cancel{BR} 0$$

$$P(X=2, Y=0) = BB = \frac{3}{8} \times \frac{2}{7} = \frac{3}{28}$$

$$P(X=2, Y=1) = 0$$

$$P(X=2, Y=2) = 0$$

© To find the Covariance of X & Y

$$\text{WKT } \sigma_{XY} = E(XY) - E(X)E(Y)$$

$$\text{Here } E(Y) = \frac{1}{2}, E(X) = \frac{3}{7}$$

$$E(XY) = \frac{3}{14}$$

$$\therefore \sigma_{XY} = \frac{3}{14} - \frac{1}{2} \times \frac{3}{7} = \frac{1}{14}$$

Marginal probabilities

Marginal probabilities of Y

X \ Y	0	1	2	$f_X(x)$
0	$\frac{3}{28}$	$\frac{6}{28}$	$\frac{1}{28}$	$\frac{10}{28}$
1	$\frac{9}{28}$	$\frac{6}{28}$	0	$\frac{15}{28}$
2	$\frac{3}{28}$	0	0	$\frac{3}{28}$
$g_Y(y)$	$\frac{18}{28}$	$\frac{12}{28}$	$\frac{1}{28}$	1

$$⑧ P(X+Y \leq 1) = \frac{18}{28}$$

$$[P(X=0, Y=0) + P(X=0, Y=1) + P(X=1, Y=0)]$$

Total probability

③ Two binomial independent RV have $p=0.3$ & $n=2$. Find the probability that the second RV is greater than the first.

X \ Y	0	1	2	$f_X(x)$
0	$(0.49)^2$	0.2058	0.0441	0.49
1	0.2058	0.1764	0.0378	0.42
2	0.0441	0.0378	$(0.09)^2$	0.09
$g_Y(y)$	0.49	0.42	0.09	1

$$(Use JPMF) \\ P(X=0) = {}^2C_0 (0.3)^0 (0.7)^2 = 0.49$$

$$P(X=1) = {}^2C_1 (0.3) (0.7) = 0.42$$

$$P(X=2) = {}^2C_2 (0.3)^2 (0.7)^0 = 0.09$$

$$P(X > Y) = 0.2058 + 0.0441 + 0.0378 = 0.2877$$

④ Let (X, Y) be a bivariate RV, where X is the input & Y is the output of binary channel. Given $P(X=0) = 0.5$, $P(Y=1|X=0) = 0.1$, $P(Y=0|X=1) = 0.2$. Find JPMF of X & Y.

X \ Y	0	1	$f_X(x)$
0	0.45	0.05	0.5
1	0.1	0.4	0.5
$g_Y(y)$	0.55	0.45	1

$$P(Y=1|X=0) = 0.1$$

$$P(Y=1, X=0) = 0.1 \times 0.5 = 0.05$$

$$P(Y=0|X=1) \Rightarrow P(Y=0, X=1) = 0.5 \times 0.2 = 0.1$$

Continuous random Variables

Let X and Y be two continuous random variables. If $f(x, y)$ is a real valued function satisfying the conditions

$$f(x, y) \geq 0 \text{ and } \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

then $f(x, y)$ is called the joint probability function or the joint density function of the random variables X and Y .

$$P(a \leq x \leq b, c \leq y \leq d) = \int_a^b \int_c^d f(x, y) dy dx$$

For any region R in 2-D space
 $P\{(X, Y) \in R\} = \iint_R f(x, y) dx dy$

Marginal probability Distributions

$P(X \leq x) = F_X(x) = \int_{-\infty}^x \int_{-\infty}^{\infty} f(u, v) dv du$ is called the ~~cumulative~~ ^{cumulative} distribution function of X .

$P(Y \leq y) = F_Y(y) = \int_{-\infty}^y \int_{-\infty}^{\infty} f(u, v) du dv$ is called the cumulative marginal distribution function of Y .

The derivative of $F_X(x)$ w.r.t x and $F_Y(y)$ w.r.t y be respectively denoted by $f_X(x)$ and $f_Y(y)$. They are given by

$$f_X(x) = \int_{-\infty}^{\infty} f(x, v) dv, \quad f_Y(y) = \int_{-\infty}^{\infty} f(u, y) du$$

$f_X(x)$ is called the marginal density function of X and $f_Y(y)$ is called the marginal density function of Y .

* The variables X and Y are said to be independent if ~~$f_X(x) \cdot f_Y(y) = f(x, y)$~~ $f_X(x) \cdot f_Y(y) = f(x, y)$.

Conditional probability density function

Given continuous random variables X and Y with joint probability density function $f_{XY}(x, y)$, the conditional probability density function of Y given $X = x$ is

$$f_{Y/X}(y) = \frac{f_{XY}(x, y)}{f_X(x)} \text{ for } f_X(x) > 0.$$

Stochastic Independence or Independent Random Variables

Let X and Y be two continuous random variables with joint probability distribution $f_{xy}(x, y)$. Then X and Y are said to be independent random variables if $f_{xy}(x, y) = f_x(x) \cdot g(y) \quad \forall x, y$

Where $f_x(x)$ is the marginal probability distribution of X
 $g(y)$ is the marginal probability distribution of Y .

Expectations and Variance

$$\begin{aligned} E(X) = \mu_x &= \int_{-\infty}^{\infty} x f_x(x) dx = \int_{-\infty}^{\infty} x \left[\int_{R_{xy}} f_{xy}(x, y) dy \right] dx \\ &= \iint_R x f_{xy}(x, y) dx dy \quad \longrightarrow (1) \end{aligned}$$

$$\begin{aligned} \text{and } V(X) = \sigma_x^2 &= \int_{-\infty}^{\infty} (x - \mu_x)^2 f_x(x) dx = \int_{-\infty}^{\infty} (x - \mu_x)^2 \left[\int_{R_{xy}} f_{xy}(x, y) dy \right] dx \\ &= \iint_R (x - \mu_x)^2 f_{xy}(x, y) dx dy \quad \longrightarrow (2) \end{aligned}$$

where R_{xy} denotes the set of all points in the range of (X, Y) for which $X=x$ and R_y denotes the set of all points in the range of (X, Y)

Definition: Let R_x denote the set of all points in the range of (X, Y) for which $X=x$. The conditional mean of Y given $X=x$, denoted as $E(Y/x)$ or $\mu_{Y/x}$, is

$$E(Y/x) = \int_{R_y} y f_{Y/x}(y) dy \quad \longrightarrow (3)$$

and the conditional variance of Y given $X=x$, denoted as $V(Y/x)$ or $\sigma_{Y/x}^2$, is

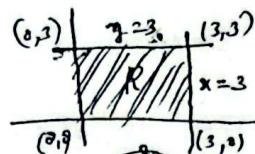
$$V(Y/x) = \int_{R_y} (y - \mu_{Y/x})^2 f_{Y/x}(y) dy = \int_{R_y} y^2 f_{Y/x}(y) dy - \mu_{Y/x}^2 \quad \longrightarrow (4)$$

Note: a) $P(a \leq x \leq b) = P(a \leq x \leq b, -\infty \leq y \leq \infty)$

b) $P(c \leq y \leq d) = P(-\infty \leq x \leq \infty, c \leq y \leq d)$

Exercises for section 5.3

- 5.34 Determine the value of c such that the function $f(x, y) = cxy$ for $0 < x < 3$ and $0 < y < 3$ satisfies the properties of a joint probability density function. Determine the following: a) $P(X < 2, Y < 3)$ b) $P(X < 2.5)$ c) $P(1 < Y < 2.5)$ d) $P(X > 1.8, 1 < Y < 2.5)$ e) $E(X)$ f) $P(X < 0, Y < 4)$ g) Marginal probability distribution of the random variable X h) Conditional probability distribution of Y given that $X = 1.5$ i) Conditional probability distribution of X given that $Y = 2$ j) $E(Y|X = 1.5)$ k) $P(X < 2|X = 1.5)$.



Sol: We must have $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$

$$\text{i.e., } \int_{y=0}^3 \int_{x=0}^3 cxy dx dy = 1 \Rightarrow c \int_{y=0}^3 y \left[\frac{x^2}{2} \right]_0^3 dy = 1 \Rightarrow \frac{c}{8} \int_{y=0}^3 y^2 dy = 1$$

$$\Rightarrow \frac{c}{8} \left[\frac{y^3}{3} \right]_0^3 = 1 \Rightarrow \frac{c}{8} \times \frac{27}{3} = 1 \Rightarrow \frac{9c}{8} = 1 \Rightarrow c = \frac{8}{9}$$

$$\text{or } \frac{9c}{8} \int_{y=0}^3 y dy = 1 \Rightarrow \frac{9c}{8} \left[\frac{y^2}{2} \right]_0^3 = 1 \Rightarrow \frac{9c}{8} \times \frac{9}{2} = 1 \Rightarrow \frac{81c}{16} = 1 \Rightarrow c = \frac{16}{81}$$

$$\text{a) } P(X < 2, Y < 3) = \int_{y=0}^3 \int_{x=0}^2 \frac{4}{81} xy dx dy = \frac{4}{81} \left[\frac{x^2}{2} \right]_0^2 \left[\frac{y^2}{2} \right]_0^3 = \frac{4}{81} \times 2 \times \frac{9}{2} = \frac{4}{9} = 0.444$$

$$\text{b) } P(X < 2.5) = \int_{y=0}^3 \int_{x=0}^{2.5} \frac{4}{81} xy dx dy = \frac{4}{81} \left[\frac{x^2}{2} \right]_0^{2.5} \left[\frac{y^2}{2} \right]_0^3 = \frac{4}{81} \times \frac{6.25}{2} \times \frac{9}{2} = \frac{6.25}{9} = 0.6944$$

$$\text{c) } P(1 < Y < 2.5) = \int_{y=1}^{2.5} \int_{x=0}^3 \frac{4}{81} xy dx dy = \frac{4}{81} \left[\frac{x^2}{2} \right]_0^3 \left[\frac{y^2}{2} \right]_1^{2.5} = \frac{4}{81} \times \frac{9}{2} \times \frac{6.25 - 1}{2} = \frac{5.25}{9} = 0.583$$

$$\text{d) } P(X > 1.8, 1 < Y < 2.5) = \int_{y=1}^{2.5} \int_{x=1.8}^3 \frac{4}{81} xy dx dy = \frac{4}{81} \left[\frac{x^2}{2} \right]_{1.8}^3 \left[\frac{y^2}{2} \right]_1^{2.5} = \frac{4}{81} \times \frac{9 - 3.24}{2} \times \frac{5.25}{2} = 0.3733$$

$$\text{e) } E(X) = \iint_R x f_{xy}(x, y) dx dy = \int_{y=0}^3 \int_{x=0}^3 x \cdot \frac{4}{81} xy dx dy = \int_{y=0}^3 \int_{x=0}^3 \frac{4}{81} x^2 y dx dy = \frac{4}{81} \left[\frac{x^3}{3} \right]_0^3 \left[\frac{y^2}{2} \right]_0^3 = \frac{4}{81} \times \frac{27}{3} \times \frac{9}{2} = 2$$

$$\text{f) } P(X < 0, Y < 4) = 0$$

$$(1) f_x(x) = \int_{y=0}^{y=x+2} f(x,y) dy = \int_{y=0}^{y=x+2} \frac{4}{81} x \cdot y dy = \frac{4}{81} x \left[\frac{y^2}{2} \right]_0^{x+2} = \frac{2}{81} x (x+2)^2 = \begin{cases} \frac{2}{9} x, & 0 < x < 3 \\ 0, & \text{otherwise} \end{cases}$$

$$(2) f_{y|x=1.5}(y) = \frac{f_{xy}(1.5, y)}{f_x(1.5)} = \frac{\frac{4}{81} \times 1.5 \times y}{\frac{2}{9} \times 1.5} = \frac{\frac{2}{27} y}{\frac{1}{3}} = \frac{2y}{9}, 0 < y < 3.$$

$$(3) f_{x|y=2}(x) = \frac{f_{xy}(x, 2)}{f_y(2)} = \frac{\frac{4}{81} \times x \times 2}{\frac{2}{9} \times 2} = \frac{\frac{8}{81} x}{\frac{4}{9}} = \frac{8}{81} x \times \frac{9}{4} = \frac{2}{9} x, \quad \left[f_y(y) = \frac{2}{9} y, 0 < y < 3 \right]$$

$$(4) E(Y|x=1.5) = \int_0^3 y \frac{2y}{9} dy = \frac{2}{9} \int_0^3 y^2 dy = \frac{2}{9} \left[\frac{y^3}{3} \right]_0^3 = \frac{2}{9} \times \frac{27}{3} = 2.$$

$$(5) P(Y < 2 | x=1.5) = \text{Probability that } y \text{ less than 2, given that } x=1.5 \text{ (The conditional density function is integrated out)}$$

$$= \int_{y=0}^2 f_{y|x=1.5}(y) dy = \int_{y=0}^2 \frac{2y}{9} dy = \frac{2}{9} \left[\frac{y^2}{2} \right]_0^2 = \frac{1}{9} \times 4 = 0.4444$$

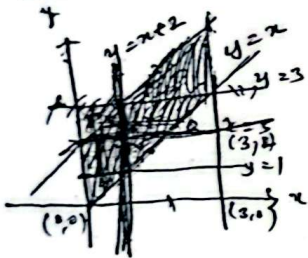
5-37 Determine the value of c that makes the function $f(x,y) = c(x+y)$ a joint probability density function over the range $0 < x < 3$ and $x < y < x+2$.

Determine the following: (a) $P(X < 1, Y < 2)$ (b) $P(1 < X < 2)$ (c) $P(Y > 1)$
(d) $P(X < 2, Y < 2)$ (e) $E(X)$

(f) Marginal probability distribution of x (g) Conditional probability distribution of y given that $x=1$ (h) $E(Y|x=1)$ (i) $P(Y > 2 | x=1)$

(j) Conditional probability distribution of x given that $y=2$:

Sol: We must have $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{xy}(x,y) = 1 \Rightarrow \int_{x=0}^3 \int_{y=x}^{x+2} c(x+y) dy dx = 1$



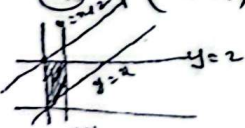
$$\Rightarrow c \int_{x=0}^3 \left[xy + \frac{y^2}{2} \right]_{y=x}^{y=x+2} dx = 1 \Rightarrow c \int_{x=0}^3 \left[x(x+2-x) + \frac{1}{2}(x^2+4x+4-x^2) \right] dx = 1$$

$$\Rightarrow c \int_{x=0}^3 (4x+2) dx = 1 \Rightarrow c \left[\frac{4x^2}{2} + 2x \right]_0^3 = 1 \Rightarrow c(18+6) = 1 \Rightarrow c = \frac{1}{24}$$

$$\therefore f(x,y) = \frac{1}{24}(x+y), 0 < x < 3, x < y < x+2$$

$$(a) P(X < 1, Y < 2) = \frac{1}{24} \int_{x=0}^1 \int_{y=x}^2 (x+y) dy dx = \frac{1}{24} \int_{x=0}^1 \left[xy + \frac{y^2}{2} \right]_{y=x}^2 dx = \frac{1}{24} \int_{x=0}^1 (2x+2) dx$$

$$= \frac{1}{24} \left[x^2 + 2x \right]_0^1 = \frac{1}{24} (1+2) = \frac{1}{8}$$



$$(b) P(1 < X < 2) = \frac{1}{24} \int_{x=1}^2 \int_{y=x}^{x+2} (x+y) dy dx = \frac{1}{24} \int_{x=1}^2 \left[xy + \frac{y^2}{2} \right]_{y=x}^{x+2} dx = \frac{1}{24} \int_{x=1}^2 (4x+2) dx$$

$$= \frac{1}{24} \left[2x^2 + 2x \right]_1^2 = \frac{1}{24} (8+4-2-1) = \frac{1}{12}$$

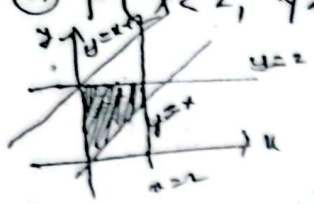


$$P(Y > 1) = \int_0^1 \int_1^2 \frac{1}{24}(x+y) dy dx = \frac{1}{24} \int_0^1 [xy + \frac{y^2}{2}]_1^2 dx = \frac{1}{24} \int_0^1 (x(2-1) + \frac{4-1}{2}) dx$$

$$= \frac{1}{24} \int_0^1 (\frac{3}{2}x + \frac{3}{2}) dx = \frac{1}{24} [\frac{3}{4}x^2 + \frac{3}{2}x]_0^1 = \frac{1}{24} (\frac{3}{4} + \frac{3}{2}) = \frac{1}{24} (\frac{3}{4} + \frac{6}{4}) = \frac{1}{24} (\frac{9}{4}) = \frac{9}{96} = \frac{3}{32}$$

$$P(X < 2, Y < 2) = \int_0^2 \int_0^2 \frac{1}{24}(x+y) dy dx = \frac{1}{24} \int_0^2 [xy + \frac{y^2}{2}]_0^2 dx = \frac{1}{24} \int_0^2 (2x + 2) dx$$

$$= \frac{1}{24} [2x^2 + 2x]_0^2 = \frac{1}{24} (8 + 4) = \frac{12}{24} = \frac{1}{2}$$

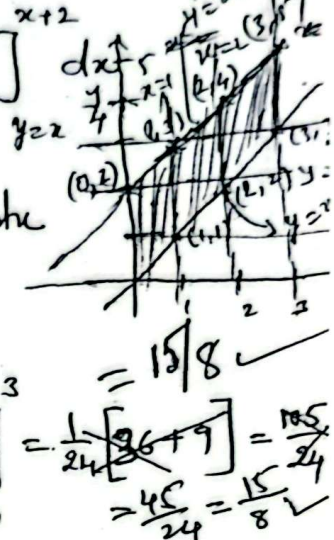


$$E(X) = \frac{1}{24} \int_0^3 \int_x^{x+2} x(x+y) dy dx = \frac{1}{24} \int_0^3 [x^2 y + x \frac{y^2}{2}]_x^{x+2} dx$$

$$= \frac{1}{24} \int_0^3 [x^2(x+2) + \frac{x}{2}(x+2)^2 - x^2 x - x \frac{x^2}{2}] dx$$

$$= \frac{1}{24} \int_0^3 (x^3 + 2x^2 + \frac{x}{2}(x^2 + 4x + 4) - x^3 - \frac{x^3}{2}) dx$$

$$= \frac{1}{24} \int_0^3 (4x^2 + 2x) dx = \frac{1}{24} [\frac{4x^3}{3} + \frac{2x^2}{2}]_0^3 = \frac{1}{24} (\frac{36}{1} + 9) = \frac{45}{24} = \frac{15}{8}$$



$$f_x(x) = \int_{y=x}^{x+2} \frac{1}{24}(x+y) dy = \frac{1}{24} [xy + \frac{y^2}{2}]_x^{x+2} = \frac{1}{24} [x(x+2-x) + \frac{(x+2)^2 - x^2}{2}]$$

$$= \frac{1}{24} [4x + 2] = \frac{2x+1}{12}, \quad 0 < x < 3$$

$$f_{Y|X=1}(y) = \frac{f_{XY}(X=1, y)}{f_X(X=1)} = \frac{\frac{1}{24}(1+y)}{\frac{3}{12}} = \frac{y+1}{6}, \quad 1 \leq y < 3$$

$$E(Y|X=1) = \int_1^3 y \left(\frac{y+1}{6}\right) dy = \frac{1}{6} \left[\frac{y^3}{3} + \frac{y^2}{2}\right]_1^3 = \frac{1}{6} \left[\frac{27}{3} - \frac{1}{3} + \frac{9}{2} - \frac{1}{2}\right] = 2.1$$

$$P(Y > 2 | X=1) = \int_2^3 f_{Y|X=1}(y) dy = \int_2^3 \frac{y+1}{6} dy = \frac{1}{6} \left[\frac{y^2}{2} + y\right]_2^3 = \frac{1}{6} \left[\frac{9}{2} + 3 - \frac{4}{2} - 2\right] = \frac{1}{6} \times \frac{7}{2} = \frac{7}{12} = 0.58$$

$$f_{X|Y=2}(x) = \frac{f_{XY}(x, Y=2)}{f_Y(Y=2)} = \frac{\frac{1}{24}(x+2)}{\int_0^3 \frac{1}{24}(x+2) dx} = \frac{\frac{1}{24}(x+2)}{\frac{1}{24} [\frac{x^2}{2} + 2x]_0^3} = \frac{2(x+2)}{15}, \quad 0 < x < 3$$

$$P(1 < X < 3, X < Y < X+2) = \frac{1}{24} \int_1^3 [xy + \frac{y^2}{2}]_x^{x+2} dx = \frac{1}{24} \int_1^3 (4x + 2) dx$$

$$= \frac{1}{24} [2x^2 + 2x]_1^3 = \frac{1}{24} (18 + 6 - 2 - 2) = \frac{10}{24} = \frac{5}{12}$$

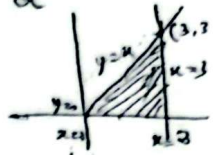
$$P(Y > 1) = \frac{68}{144} + \frac{1}{144} = \frac{69}{144} = \frac{23}{48}$$

(5.40) Determine the value of c that makes the function $f(x, y) = cxy$ a joint probability density function over the range $0 < x < 3$ and $0 < y < x$.

Sol: We must have $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$

$$\text{i.e. } \int_{x=0}^3 \int_{y=0}^x cxy dy dx = 1 \quad \text{i.e. } c \int_{x=0}^3 x \left[\frac{y^2}{2} \right]_0^x dx = 1 \quad \text{i.e. } c \int_{x=0}^3 \frac{x^3}{2} dx = 1$$

$$\text{i.e. } \frac{c}{8} [x^4]_0^3 = 1 \quad \text{i.e. } \frac{c}{8} [81] = 1 \quad \text{i.e. } c = \frac{8}{81}$$



(5.41) Continuation of Exercise 5.40. Determine the following:

- (a) $P(X < 1, Y < 2)$ (b) $P(1 < X < 2)$ (c) $P(Y > 1)$ (d) $P(X < 2, Y < 2)$ (e) $E(X), E(Y)$

Sol: (a) $P(X < 1, Y < 2) = \int_0^1 \int_0^x \frac{8}{81} xy dy dx = \frac{8}{81} \int_0^1 x \left[\frac{y^2}{2} \right]_0^x dx = \frac{8}{81} \int_0^1 \frac{x^3}{2} dx = \frac{8}{81} \cdot \frac{1}{8} = \frac{1}{81}$

(b) $P(1 < X < 2) = \int_1^2 \int_0^x \frac{8}{81} xy dy dx = \frac{8}{81} \int_1^2 x \left[\frac{y^2}{2} \right]_0^x dx = \frac{8}{81} \int_1^2 \frac{x^3}{2} dx = \frac{8}{81} \cdot \frac{1}{8} (16 - 1) = \frac{15}{81} = \frac{5}{27}$

(c) $P(Y > 1) = \int_1^3 \int_1^x \frac{8}{81} xy dy dx = \frac{8}{81} \int_1^3 x \left[\frac{y^2}{2} \right]_1^x dx = \frac{8}{81} \int_1^3 x \left(\frac{x^2}{2} - \frac{1}{2} \right) dx = \frac{4}{81} \int_1^3 (x^3 - 1) dx = \frac{4}{81} \left[\frac{x^4}{4} - x \right]_1^3 = \frac{4}{81} \left(\frac{81}{4} - 3 - \frac{1}{4} + 1 \right) = \frac{4}{81} \left(\frac{80}{4} - 2 \right) = \frac{4}{81} \cdot 18 = \frac{72}{81} = \frac{8}{9}$

(d) $P(X < 2, Y < 2) = \int_0^2 \int_0^x \frac{8}{81} xy dy dx = \frac{8}{81} \int_0^2 x \left[\frac{y^2}{2} \right]_0^x dx = \frac{8}{81} \int_0^2 \frac{x^3}{2} dx = \frac{8}{81} \cdot \frac{1}{8} (16 - 0) = \frac{16}{81}$

(e) $E(X) = \int_0^3 \int_0^x x \left(\frac{8}{81} xy \right) dy dx = \frac{8}{81} \int_0^3 x \left[\frac{y^2}{2} \right]_0^x dx = \frac{8}{81} \int_0^3 \frac{x^3}{2} dx = \frac{4}{81} \int_0^3 x^3 dx = \frac{4}{81} \left[\frac{x^4}{4} \right]_0^3 = \frac{4}{81} \cdot \frac{81}{4} = 1$

(5.42) Continuation of Exercise 5.40. Determine the following:

- (a) Marginal probability distribution of X (b) Conditional probability distribution of Y given $X=1$ (c) $E(Y|X=1)$ (d) $P(Y > 2|X=1)$ (e) Conditional probability of X given $Y=2$.

Solution: (a) $P(X < 1, Y < 2) = \int_{x=0}^1 \int_{y=0}^x \frac{8}{81} xy dy dx = \frac{8}{81} \int_{x=0}^1 x \left[\frac{y^2}{2} \right]_0^x dx = \frac{8}{81} \int_{x=0}^1 \frac{x^3}{2} dx = \frac{8}{81} \cdot \frac{1}{8} = \frac{1}{81}$

(b) $P(1 < X < 2) = \int_{x=1}^2 \int_{y=0}^x \frac{8}{81} xy dy dx = \frac{8}{81} \left[\frac{x^4}{8} \right]_1^2 = \frac{5}{27}$

(c) $P(Y > 1) = \int_{y=1}^3 \int_{x=y}^3 \frac{8}{81} xy dx dy = \frac{64}{81}$

(d) $P(X < 2, Y < 2) = \int_0^2 \int_0^x \frac{8}{81} xy dy dx = \frac{16}{81}$

5.43 Determine the value of c that makes the function $f(x,y) = ce^{-2x-3y}$ a joint probability density function over the range $x > 0$ and $0 < y < x$.

Sol: $\int_0^\infty \int_0^x ce^{-2x-3y} dy dx = 1 \Rightarrow c \int_0^\infty \int_0^x e^{-2x} \cdot e^{-3y} dy dx = 1$

$\Rightarrow c \int_0^\infty e^{-2x} \left[\frac{e^{-3y}}{-3} \right]_0^x dx = 1 \Rightarrow c \int_0^\infty \frac{e^{-2x}}{-3} (e^{-3x} - 1) dx = 1$

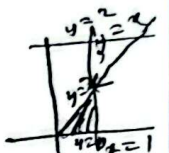
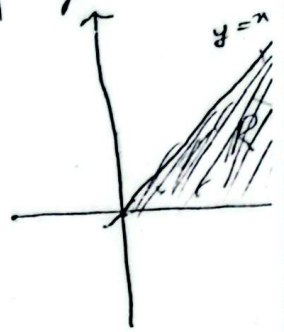
$\Rightarrow \frac{c}{3} \int_0^\infty (e^{-2x} - e^{-5x}) dx = 1 \Rightarrow \frac{c}{3} \left[\frac{e^{-2x}}{-2} - \frac{e^{-5x}}{-5} \right]_0^\infty = 1$

$\Rightarrow \frac{c}{3} \left[\frac{1}{5} (0 - 1) - \frac{1}{2} (0 - 1) \right] = 1$

$\Rightarrow \frac{c}{3} \left[\frac{1}{5} (-1) - \frac{1}{2} (-1) \right] = 1$

$\Rightarrow \frac{c}{3} \left(\frac{1}{2} - \frac{1}{5} \right) = 1 \Rightarrow \frac{c}{3} \times \frac{5-2}{10} = 1 \Rightarrow c = 10$

$\therefore f(x,y) = 10e^{-2x-3y}, x > 0, 0 < y < x$



5.44 continuation of Exercise 5.43. Determine the following:

(a) $P(X < 1, Y < 2)$ (b) $P(1 < X < 2)$ (c) $P(Y > 3)$ (d) $P(X < 2, Y < 2)$

(e) $E(X)$ (f) $E(Y)$

Sol: (a) $P(X < 1, Y < 2) = \int_0^1 \int_0^x 10e^{-2x-3y} dy dx = 10 \int_0^1 \left[\frac{e^{-3y}}{-3} \right]_0^x dx = \frac{10}{3} \int_0^1 (1 - e^{-3x}) dx = \frac{10}{3} \left[x - \frac{e^{-3x}}{-3} \right]_0^1 = \frac{10}{3} \left(1 - \frac{1}{3} (1 - e^{-3}) \right) = \frac{10}{3} \left(\frac{2}{3} + \frac{e^{-3}}{3} \right) = \frac{10}{9} (2 + e^{-3}) \approx 2.77$

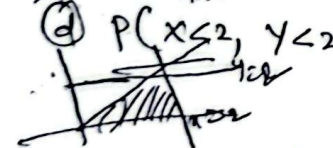
(b) $P(1 < X < 2) = \int_1^2 \int_0^x 10e^{-2x-3y} dy dx = \int_1^2 10e^{-2x} \left[\frac{e^{-3y}}{-3} \right]_0^x dx = \frac{10}{3} \int_1^2 (1 - e^{-3x}) dx = \frac{10}{3} \left[x - \frac{e^{-3x}}{-3} \right]_1^2 = \frac{10}{3} \left(2 - \frac{1}{3} (e^{-6} - e^{-3}) \right) = \frac{10}{3} \left(\frac{5}{3} + \frac{e^{-3}}{3} - \frac{e^{-6}}{3} \right) = \frac{10}{9} (5 + e^{-3} - e^{-6})$



(c) $P(Y > 3) = \int_3^\infty \int_x^\infty 10e^{-2x-3y} dy dx = \int_3^\infty 10e^{-2x} \left[\frac{e^{-3y}}{-3} \right]_x^\infty dx = \frac{10}{3} \int_3^\infty e^{-2x} (e^{-3x} - 1) dx = \frac{10}{3} \int_3^\infty (e^{-5x} - e^{-2x}) dx = \frac{10}{3} \left[\frac{e^{-5x}}{-5} - \frac{e^{-2x}}{-2} \right]_3^\infty = \frac{10}{3} \left(\frac{1}{2} e^{-6} - \frac{1}{5} e^{-3} \right) = \frac{10}{3} \left(\frac{5e^{-6} - 2e^{-3}}{10} \right) = \frac{1}{3} (5e^{-6} - 2e^{-3})$



(d) $P(X < 2, Y < 2) = \int_0^2 \int_0^x 10e^{-2x-3y} dy dx = \frac{10}{3} \int_0^2 (1 - e^{-3x}) dx = \frac{10}{3} \left[x - \frac{e^{-3x}}{-3} \right]_0^2 = \frac{10}{3} \left(2 - \frac{1}{3} (e^{-6} - 1) \right) = \frac{10}{3} \left(\frac{5}{3} + \frac{1}{3} - \frac{e^{-6}}{3} \right) = \frac{10}{9} (6 - e^{-6})$



(e) $E(X) = \int_0^\infty \int_0^x 10x e^{-2x-3y} dy dx = \int_0^\infty 10x e^{-2x} \left[\frac{e^{-3y}}{-3} \right]_0^x dx = \frac{10}{3} \int_0^\infty x (1 - e^{-3x}) dx = \frac{10}{3} \left[\int_0^\infty x dx - \int_0^\infty x e^{-3x} dx \right] = \frac{10}{3} \left[\frac{x^2}{2} - \left(\frac{x e^{-3x}}{-3} - \frac{e^{-3x}}{9} \right) \right]_0^\infty = \frac{10}{3} \left(\frac{\infty}{2} - \left(-\frac{\infty}{3} - \frac{1}{9} \right) \right) = \frac{10}{3} \left(\frac{\infty}{2} + \frac{\infty}{3} + \frac{1}{9} \right) = \frac{10}{3} \left(\frac{5\infty}{6} + \frac{1}{9} \right) = \frac{10}{3} \left(\frac{5\infty}{6} \right) = \frac{50\infty}{6} = \frac{25\infty}{3}$

$= 5 \int_0^\infty x e^{-2x} dx - 5 \int_0^\infty x e^{-4x} dx = 5 \left[\frac{x e^{-2x}}{-2} - \frac{e^{-2x}}{4} - \left(\frac{x e^{-4x}}{-4} - \frac{e^{-4x}}{16} \right) \right]_0^\infty = 5 \left[\frac{1}{4} - \frac{1}{16} \right] = 5 \times \frac{3}{16} = \frac{15}{16}$

(f) $E(Y) = \int_0^\infty \int_0^x 10e^{-2x-3y} y dy dx = 10 \int_0^\infty e^{-2x} \left[\frac{y e^{-3y}}{-3} - \frac{e^{-3y}}{9} \right]_0^x dx = 10 \int_0^\infty e^{-2x} \left(-\frac{x e^{-3x}}{3} - \frac{e^{-3x}}{9} \right) dx = 10 \int_0^\infty \left(-\frac{x e^{-5x}}{3} - \frac{e^{-5x}}{9} \right) dx = 10 \left[-\frac{x e^{-5x}}{15} - \frac{e^{-5x}}{45} \right]_0^\infty = 10 \left(\frac{1}{45} \right) = \frac{2}{9}$

5.45 Continuation of Exercise 5.43. Determine the following:

(a) Marginal probability distribution of X (b) Conditional probability distribution of Y given $X=1$

(c) $E(Y|X=1)$ (d) Conditional probability distribution of X given $Y=2$.

2) (a) $f_X(x) = \int_{y=0}^x 10 e^{-2x} e^{-3y} dy = 10 e^{-2x} \left[\frac{e^{-3y}}{-3} \right]_0^x = \frac{10}{3} e^{-2x} (1 - e^{-3x}) = \frac{10}{3} (e^{-2x} - e^{-5x})$ for $0 < x$

(b) $f_{Y|X=1}(y) = \frac{f_{XY}(1, y)}{f_X(x=1)} = \frac{10 e^{-2} e^{-3y}}{\frac{10}{3} (e^{-2} - e^{-5})} = 3 e^{-3y} \times \frac{1}{1 - e^{-3}} = 3.1572 e^{-3y}, \text{ for } y > 0$

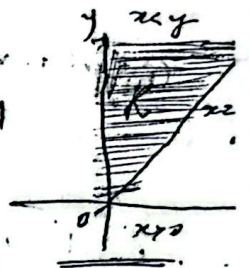
(c) $E(Y|X=1) = \int_0^1 y \times f_{Y|X=1}(y) dy = \int_0^1 3.1572 y e^{-3y} dy$
 $= 3.1572 \left[y \frac{e^{-3y}}{-3} + \frac{e^{-3y}}{9} \right]_0^1 = 3.1572 \left[\frac{e^{-3}}{-3} + \frac{e^{-3}}{9} \right] = 3.1572 \left[\frac{e^{-3}}{9} - \frac{e^{-3}}{3} \right] = 3.1572 \left[\frac{e^{-3}}{9} (1 - 3) \right] = 3.1572 \left[\frac{e^{-3}}{9} (-2) \right] = -0.771$

(d) $f_{X|Y=2}(x) = \frac{f_{XY}(x, 2)}{f_Y(y=2)}$
 $= \frac{10 e^{-2x} e^{-6}}{5 e^{-6}} = 2 e^{-2x}, 2 < x$
 $f_Y(y) = \int_{x=0}^{\infty} 10 e^{-2x} e^{-6} dx = 10 e^{-6} \left[\frac{e^{-2x}}{-2} \right]_0^{\infty} = 5 e^{-6} (1 - 0) = 5 e^{-6}$

5.46 Determine the value of c that makes the function $f(x, y) = c e^{-2x - 3y}$ a joint probability density function over the range $0 < x$ and $x < y$.

Determine the questions given in Ex. 44 & 5.45.

$\int_0^{\infty} \int_{y=x}^{\infty} c e^{-2x} e^{-3y} dy dx = 1 \Rightarrow c \int_0^{\infty} e^{-2x} \left[\frac{e^{-3y}}{-3} \right]_{y=x}^{\infty} dx = 1$
 $\Rightarrow \frac{c}{3} \int_0^{\infty} e^{-2x} (e^{-3x} - 0) dx = 1 \Rightarrow \frac{c}{3} \left[\frac{e^{-5x}}{-5} \right]_0^{\infty} = 1$
 $\Rightarrow \frac{c}{15} = 1 \Rightarrow c = 15$



5.47 The joint pdf of (X, Y) is given by $f(x, y) = 24xy$ $x > 0, y > 0, x + y \leq 1$, and $f(x, y) = 0$, elsewhere, find the conditional mean and variance of Y , given X .

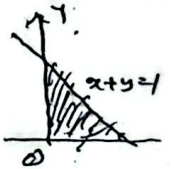
So: $f_X(x) = \int_0^{1-x} 24xy dy = 12x(1-x)^2, 0 < x < 1$

Now $f_{Y|X}(y) = \frac{f(x, y)}{f_X(x)} = \frac{24y}{(1-x)^2}, 0 < y < 1-x$

$E(Y|X=x) = \int_0^{1-x} y f_{Y|X}(y) dy = \int_0^{1-x} \frac{24y^2}{(1-x)^2} dy = \frac{2}{3} \left[\frac{y^3}{(1-x)^2} \right]_0^{1-x} = \frac{2}{3}(1-x)$

$E(Y^2|X=x) = \int_0^{1-x} y^2 f_{Y|X}(y) dy = \int_0^{1-x} \frac{24y^3}{(1-x)^2} dy = \frac{1}{2}(1-x)^2$

Variance of $(Y|X=x) = E(Y^2|X=x) - [E(Y|X=x)]^2 = \frac{1}{2}(1-x)^2 - \frac{4}{9}(1-x)^2 = \frac{1}{18}(1-x)^2$



(2) The joint pdf of a two-dimensional RV (X, Y) is given by

$$f(x, y) = xy^2 + \frac{x^2}{8}, \quad 0 \leq x \leq 2, \quad 0 \leq y \leq 1$$

compute $P(X > 1)$, $P(Y < \frac{1}{2})$, $P(X > 1 | Y < \frac{1}{2})$, $P(Y < \frac{1}{2} | X > 1)$, $P(X < Y)$ and $P(X + Y \leq 1)$.

Sol Here the rectangle defined by $0 \leq x \leq 2$, $0 \leq y \leq 1$ in the range space $R_1, R_2, \dots$ are events spaces.

$$(i) P(X > 1) = \iint_{R_1(X > 1)} f(x, y) dx dy = \int_1^2 \int_0^1 (xy^2 + \frac{x^2}{8}) dy dx = \frac{19}{24}$$

$$(ii) P(Y < \frac{1}{2}) = \iint_{R_2(Y < \frac{1}{2})} (xy^2 + \frac{x^2}{8}) dx dy = \int_0^{\frac{1}{2}} \int_0^2 (xy^2 + \frac{x^2}{8}) dx dy = \frac{1}{4}$$

$$(iii) P(X > 1, Y < \frac{1}{2}) = \iint_{R_3(X > 1, Y < \frac{1}{2})} (xy^2 + \frac{x^2}{8}) dx dy = \int_0^{\frac{1}{2}} \int_1^2 (xy^2 + \frac{x^2}{8}) dx dy = \frac{5}{24}$$

$$(iv) P(X > 1 | Y < \frac{1}{2}) = \frac{P(X > 1, Y < \frac{1}{2})}{P(Y < \frac{1}{2})} = \frac{5/24}{1/4} = \frac{5}{6}$$

$$(v) P(Y < \frac{1}{2} | X > 1) = \frac{P(X > 1, Y < \frac{1}{2})}{P(X > 1)} = \frac{5/24}{19/24} = \frac{5}{19}$$

$$(vi) P(X < Y) = \iint_{R_4(X < Y)} (xy^2 + \frac{x^2}{8}) dx dy = \int_0^1 \int_0^y (xy^2 + \frac{x^2}{8}) dx dy = \frac{53}{480}$$

$$(vii) P(X + Y \leq 1) = \iint_{R_5(X + Y \leq 1)} (xy^2 + \frac{x^2}{8}) dx dy = \int_0^1 \int_0^{1-y} (xy^2 + \frac{x^2}{8}) dx dy = \frac{13}{480}$$

(3) The joint pdf of the RV (X, Y) is given by

$f(x, y) = kxy e^{-(x^2 + y^2)}$, $x > 0, y > 0$. Find the value of k and prove that X and Y are independent.

Here the range space is the entire first quadrant of the xy plane.

Sol By the property of the joint pdf $\int \int kxy e^{-(x^2 + y^2)} dx dy = 1$

$$\therefore k \int_0^{\infty} y e^{-y^2} dy \int_0^{\infty} x e^{-x^2} dx = 1 \Rightarrow \frac{k}{4} = 1 \therefore k = 4$$

$$\begin{aligned} \text{Now, } f_x(x) &= \int_{y=0}^{\infty} f(x,y) dy = 4 \int_{y=0}^{\infty} x e^{-x^2} \times y e^{-y^2} dy = 4x e^{-x^2} \int_{y=0}^{\infty} y e^{-y^2} dy \\ &= 4x e^{-x^2} \int_{t=0}^{\infty} e^{-t} \frac{dt}{2} \quad \text{where } y^2 = t, 2y dy = dt, y dy = \frac{dt}{2} \\ &= 2x e^{-x^2} \left[-e^{-t} \right]_0^{\infty} = 2x e^{-x^2}, \quad x > 0 \end{aligned}$$

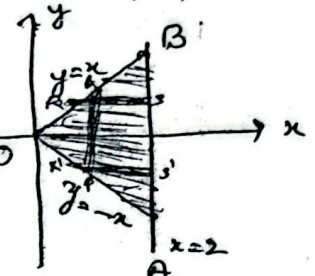
$$\text{Similarly } f_y(y) = 2y e^{-y^2}, \quad y > 0$$

$$\text{Now } f_x(x) \times f_y(y) = 4xy e^{-(x^2+y^2)} = f(x,y)$$

$\therefore$ the random variables x and y are independent.

- ④ Given $f_{xy}(x,y) = cx(x-y)$, $0 < x < 2$, $-x < y < x$, and 0 elsewhere,
 (a) evaluate c , (b) find $f_x(x)$, (c) $f_{y|x}(y|x)$ and (d) $f_y(y)$

Sol Here the range space is the area within the triangle OAB (shown in the figure) defined by $0 < x < 2$ and $-x < y < x$.



(a) By the property of pdf

$$\begin{aligned} \iint_{\Delta OAB} cx(x-y) dx dy &= 1 \quad \therefore \int_{x=0}^2 \int_{y=-x}^x cx(x-y) dy dx = 1 \\ \therefore c \int_0^2 \left[x^2 y - x \frac{y^2}{2} \right]_{-x}^x dx &= 1 \quad \therefore c \int_0^2 \left(x^3 - \frac{x^3}{2} + x^3 - \frac{x^3}{2} \right) dx = 1 \\ \therefore c \left[\frac{2x^4}{4} \right]_0^2 &= 1 \quad \therefore 8c = 1 \Rightarrow c = \frac{1}{8} \end{aligned}$$

$$(b) f_x(x) = \int_{-x}^x \frac{1}{8} x(x-y) dy = \frac{1}{8} \left[x^2 y - x \frac{y^2}{2} \right]_{-x}^x = \frac{1}{8} \left[x^3 - \frac{x^3}{2} + x^3 - \frac{x^3}{2} \right] = \frac{x^2}{4}, \quad 0 < x < 2$$

$$(c) f_{y|x}(y) = \frac{f_{xy}(x,y)}{f_x(x)} = \frac{\frac{1}{8} x(x-y)}{\frac{x^2}{4}} = \frac{1}{2x} (x-y), \quad -x < y < x$$

$$\begin{aligned} (d) f_y(y) &= \begin{cases} \int_{-y}^y \frac{1}{8} x(x-y) dx, & \text{in } -2 \leq y \leq 0 \\ \int_y^2 \frac{1}{8} x(x-y) dx, & \text{in } 0 \leq y \leq 2 \end{cases} \\ &= \begin{cases} \frac{1}{8} \left[\frac{x^2}{2} xy \right]_{-y}^y = \frac{1}{8} \left[\frac{y^2}{2} - \frac{y^2}{2} \right] = 0, & \text{in } -2 \leq y \leq 0 \\ \frac{1}{8} \left[\frac{x^2}{2} - 2xy + \frac{y^2}{2} \right]_y^2 = \frac{1}{8} \left[\frac{4}{2} - 4y + \frac{y^2}{2} - \left(\frac{y^2}{2} - 2y^2 + \frac{y^2}{2} \right) \right] \\ = \frac{1}{8} \left[2 - 4y + \frac{y^2}{2} + 2y^2 - \frac{y^2}{2} \right] = \frac{1}{8} (2 - 4y + 2y^2), & \text{in } 0 \leq y \leq 2 \end{cases} \\ \therefore f_y(y) &= \begin{cases} \frac{1}{8} (2 - 4y + 2y^2), & \text{in } 0 \leq y \leq 2 \\ 0, & \text{in } -2 \leq y \leq 0 \end{cases} \end{aligned}$$

COVARIANCE AND CORRELATION

The covariance between the random variables X and Y , denoted as $\text{cov}(X, Y)$ or σ_{xy} , is
 $\rightarrow$ how two variables are related.

$$\sigma_{xy} = E[(X - \mu_x)(Y - \mu_y)] = E(XY) - \mu_x \mu_y$$

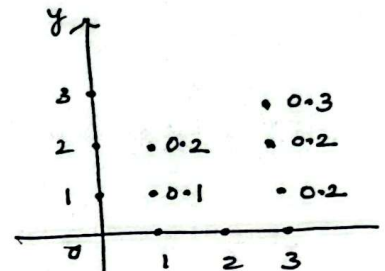
The correlation between the random variables X and Y , denoted as ρ_{xy} , is $\rho_{xy} = \frac{\text{cov}(X, Y)}{\sqrt{V(X)}\sqrt{V(Y)}} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$

For any two random variables X and Y , $-1 \leq \rho_{xy} \leq +1$
 Note: The covariance is defined for both continuous and discrete RVs by the same formula
Examples:

① For the joint probability distribution of the two random variables in the below figure, calculate $E[(X - \mu_x)(Y - \mu_y)]$

Sol: $\mu_x = 1 \times 0.3 + 3 \times 0.7 = 2.4$

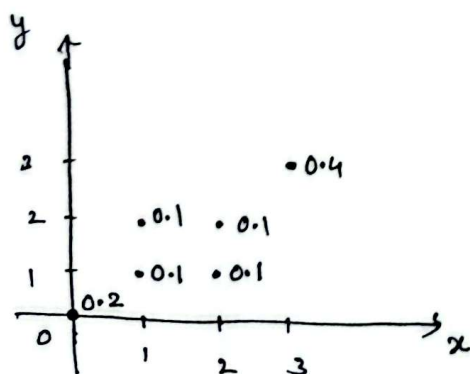
and $\mu_y = 1 \times 0.3 + 2 \times 0.4 + 3 \times 0.3 = 2.0$



Therefore,

$$\begin{aligned} E[(X - \mu_x)(Y - \mu_y)] &= \sum_{x,y} (x - \mu_x)(y - \mu_y) f_{xy}(x, y) \\ &= (1 - 2.4)(1 - 2.0) \times 0.1 + (1 - 2.4)(2 - 2.0) \times 0.2 \\ &\quad + (3 - 2.4)(1 - 2.0) \times 0.2 + (3 - 2.4)(2 - 2.0) \times 0.2 \\ &\quad + (3 - 2.4)(3 - 2.0) \times 0.3 = 0.2 \end{aligned}$$

② For the discrete random variables X and Y with the joint distribution shown in figure, determine σ_{xy} and ρ_{xy} .



The calculations for $E(XY)$, $E(X)$, and $E(Y)$ are as follows:

$$E(XY) = 0 \times 0 \times 0.2 + 1 \times 1 \times 0.1 + 1 \times 2 \times 0.1 + 2 \times 1 \times 0.1 + 2 \times 2 \times 0.1 + 3 \times 3 \times 0.4 = 4.5$$

$$E(X) = 0 \times 0.2 + 1 \times 0.2 + 2 \times 0.2 + 3 \times 0.4 = 1.8$$

$$V(X) = (0 - 1.8)^2 \times 0.2 + (1 - 1.8)^2 \times 0.2 + (2 - 1.8)^2 \times 0.2 + (3 - 1.8)^2 \times 0.4 = 1.36$$

Because the marginal probability distribution of Y is the same as for X , $E(Y) = 1.8$ and $V(Y) = 1.36$. Consequently,

$$\sigma_{xy} = E(XY) - E(X)E(Y) = 4.5 - (1.8)(1.8) = 1.26$$

Furthermore,
$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} = \frac{1.26}{(\sqrt{1.36})(\sqrt{1.36})} = 0.926$$

If X and Y are independent random variables, $\sigma_{xy} = \rho_{xy} = 0$

③ For the jpdf of two random variables X and Y , $f_{xy}(x, y) = \frac{1}{16}xy$
 $0 \leq x \leq 4, 0 \leq y \leq 2$
 show that $\sigma_{xy} = 0$.

Sol. The two random variables in this example are continuous random variables. In this case $E(XY)$ is defined as the double integral over the range of (X, Y) . That is,

$$\begin{aligned} E(XY) &= \int_0^4 \int_0^2 xy f_{xy}(x, y) dx dy = \frac{1}{16} \int_0^4 \left[\int_0^2 x^2 y^2 dx \right] dy \\ &= \frac{1}{16} \left[\frac{x^3}{3} \right]_0^2 \left[\frac{y^3}{3} \right]_0^4 = \frac{1}{16} \times \frac{8}{3} \times \frac{64}{3} = \frac{32}{9} \end{aligned}$$

$$\begin{aligned} \text{Also, } E(X) &= \int_0^4 \int_0^2 x f_{xy}(x, y) dx dy = \frac{1}{16} \int_0^4 \left[\int_0^2 x^2 dx \right] y dy \\ &= \frac{1}{16} \left[\frac{x^3}{3} \right]_0^2 \left[\frac{y^2}{2} \right]_0^4 = \frac{1}{16} \times \frac{8}{3} \times \frac{16}{2} = \frac{4}{3} \end{aligned}$$

$$E(Y) = \int_0^4 \int_0^2 y f_{xy}(x, y) dx dy = \frac{1}{16} \left[\frac{x^2}{2} \right]_0^2 \left[\frac{y^3}{3} \right]_0^4 = \frac{1}{16} \times \frac{4}{2} \times \frac{64}{3} = \frac{8}{3}$$

$$\therefore \text{Thus, } E(XY) - E(X)E(Y) = \frac{32}{9} - \frac{32}{9} = 0$$

It can be shown that these two random variables are independent. You can check that $f_{xy}(x, y) = f_x(x) f_y(y)$ for all x, y .

However, if the correlation between two random variables is zero we cannot immediately conclude that the random variables are independent.

Exercises for Section 5.5

5.67 Determine the covariance and correlation for the following joint probability distribution:

x	1	1	2	4
y	3	4	5	6
$f_{xy}(x,y)$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{8}$

can rewrite

Sol: Joint probability distribution of X and Y is as follows:

$x \backslash y$	3	4	5	6	$f_x(x)$
1	$\frac{1}{8}$	$\frac{1}{4}$	0	0	$\frac{3}{8}$
2	0	0	$\frac{1}{2}$	0	$\frac{1}{2}$
4	0	0	0	$\frac{1}{8}$	$\frac{1}{8}$
$g_y(y)$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{8}$	1

The calculations for $E(XY)$, $E(X)$, $E(Y)$ and $V(X)$, $V(Y)$ are as follows:

$$E(XY) = 1 \times 3 \times \frac{1}{8} + 1 \times 4 \times \frac{1}{4} + 1 \times 5 \times 0 + 1 \times 6 \times 0 + 2 \times 3 \times 0 + 2 \times 4 \times 0 + 2 \times 5 \times \frac{1}{2} + 2 \times 6 \times 0 + 4 \times 3 \times 0 + 4 \times 4 \times 0 + 4 \times 5 \times 0 + 4 \times 6 \times \frac{1}{8}$$

$$= \frac{3}{8} + 1 + 5 + 3 = \frac{3}{8} + 9 = \frac{75}{8}$$

$$E(X) = 1 \times \frac{3}{8} + 2 \times \frac{1}{2} + 4 \times \frac{1}{8} = \frac{3}{8} + 1 + \frac{1}{2} = \frac{3}{8} + \frac{3}{2} = \frac{15}{8}$$

$$E(Y) = 3 \times \frac{1}{8} + 4 \times \frac{1}{4} + 5 \times \frac{1}{2} + 6 \times \frac{1}{8} = \frac{3}{8} + 1 + \frac{5}{2} + \frac{6}{8} = \frac{37}{8}$$

$$\therefore \text{Cov}(X, Y) = \sigma_{xy} = E(XY) - E(X) \cdot E(Y) = \frac{75}{8} - \frac{15}{8} \times \frac{37}{8} = \frac{75 \times 8 - 15 \times 37}{64}$$

$$= \frac{600 - 555}{64} = \frac{45}{64} = 0.7031$$

$$V(X) = 1 \times \frac{3}{8} + 4 \times \frac{1}{2} + 16 \times \frac{1}{8} - \left(\frac{15}{8}\right)^2 = \frac{3}{8} + 2 + 2 - \frac{225}{64} = \frac{35}{8} - \frac{225}{64} = \frac{280 - 225}{64} = \frac{55}{64}$$

$$V(Y) = 9 \times \frac{1}{8} + 16 \times \frac{1}{4} + 25 \times \frac{1}{2} + 36 \times \frac{1}{8} - \left(\frac{37}{8}\right)^2 = \frac{9}{8} + \frac{32}{4} + \frac{100}{2} + \frac{36}{8} - \frac{1369}{64} = \frac{177}{8} - \frac{1369}{64} = \frac{1416 - 1369}{64} = \frac{47}{64}$$

$$\therefore \rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} = \frac{\frac{45}{64}}{\sqrt{\frac{55}{64}} \sqrt{\frac{47}{64}}} = \frac{45}{\sqrt{55} \sqrt{47}} = \frac{45}{\sqrt{2585}} = \frac{45}{50.84} = 0.885$$

(2) Determine the covariance and correlation for the following joint probability distribution:

x	-1	-0.5	0.5	1
y	-2	-1	1	2
$f_{xy}(x,y)$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{8}$

(5.69) Determine the value of c and the covariance and correlation for the joint probability mass function $f_{xy}(x,y) = c(x+y)$ for $x=1,2,3$ and $y=1,2,3$.

Sol, joint probability mass function of x and y is

$x \backslash y$	1	2	3	$f(x_i)$
1	$2c$	$3c$	$4c$	$9c$
2	$3c$	$4c$	$5c$	$12c$
3	$4c$	$5c$	$6c$	$15c$
$g(y_j)$	$9c$	$12c$	$15c$	$36c$

$$\text{we must have } \sum_{x=1}^3 \sum_{y=1}^3 f_{xy}(x,y) = 1$$

$$36c = 1 \Rightarrow c = \frac{1}{36}$$

$$\therefore f_{xy}(x,y) = \frac{1}{36}(x+y), \quad x=1,2,3, \quad y=1,2,3.$$

$$\begin{aligned} E(xy) &= 1 \times 1 \times 2c + 1 \times 2 \times 3c + 1 \times 3 \times 4c + 2 \times 1 \times 3c + 2 \times 2 \times 4c + 2 \times 3 \times 5c \\ &\quad + 3 \times 1 \times 4c + 3 \times 2 \times 5c + 3 \times 3 \times 6c = 2c + 6c + 12c + 6c + 16c + 30c \\ &\quad + 12c + 30c + 54c = 168c = \frac{168}{36} = \frac{14}{3} \end{aligned}$$

$$E(x) = 1 \times 9c + 2 \times 12c + 3 \times 15c = 78c = \frac{78}{36} = \frac{13}{6} = E(y)$$

$$E(x^2) = 1 \times 9c + 4 \times 12c + 9 \times 15c = 190c = \frac{190}{36} = \frac{95}{18} = E(y^2)$$

$$V(x) = V(y) = \frac{95}{18} - \left(\frac{13}{6}\right)^2 = \frac{95 \times 18 - 39 \times 39}{18 \times 18} = \frac{189}{324} = \frac{7}{18}$$

$$\therefore \sigma_{xy} = E(xy) - E(x) \cdot E(y) = \frac{14}{3} - \frac{13}{6} \times \frac{13}{6} = \frac{84 \times 18 - 39^2}{18 \times 18} = \frac{-9}{18 \times 18}$$

$$\text{and } \rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} = \frac{-1/36}{\sqrt{\frac{7}{18}} \times \sqrt{\frac{7}{18}}} = \frac{-1/36}{\frac{7}{18}} = -\frac{1}{36} \times \frac{18}{7} = -\frac{1}{14} \approx -0.0714$$

(8.72) Determine the value ~~for~~ for c and the covariance and correlation for the joint probability density function $f_{xy}(x,y)=c$ over the range $0 < x < 3$ and $0 < y < x$.

(5.73) Determine the value for c and the covariance and correlation for the joint probability density function $f_{xy}(x,y)=c$ over the range $0 < x < 5$, $0 < y$, and $x-1 < y < x+1$.

Ans: $c = 9.5$, $\sigma_{xy} = 1.852$, $\rho_{xy} = 0.928$.

(5.75) Determine the covariance and correlation for the joint probability density function $f_{xy}(x,y) = e^{-x-y}$ over the range $0 < x$ and $0 < y$.

Sol: x and y are independent and $\sigma_{xy} = 0 \Rightarrow \rho_{xy} = 0$.

(5.77) The joint probability distribution is

x	:	-1	0	0	1
y	:	0	-1	1	0
$f_{xy}(x,y)$:		$1/4$	$1/4$	$1/4$	$1/4$

Show that the correlation between x and y is zero, but x and y are not independent.

Answers: (5.75)

we must have $\int_0^\infty \int_0^\infty e^{-x} e^{-y} dx dy = 1$
 $\Rightarrow \left[e^{-x} \right]_0^\infty \left[e^{-y} \right]_0^\infty = 1$

$$\begin{aligned} E(xy) &= \int_0^\infty \int_0^\infty xy e^{-x} e^{-y} dx dy = \int_0^\infty x e^{-x} dx \times \int_0^\infty y e^{-y} dy \\ &= \left[\frac{x e^{-x}}{-1} - 1 \cdot \frac{e^{-x}}{-1} \right]_0^\infty \times \left[\frac{y e^{-y}}{-1} - \frac{e^{-y}}{-1} \right]_0^\infty \\ &= 1 \times 1 = 1 \end{aligned}$$

$$\text{Also, } E(X) = \int_0^{\infty} \int_0^{\infty} x f_{xy}(u, y) dx dy = \int_0^{\infty} \int_0^{\infty} x e^{-x} e^{-y} dx dy$$

$$= \left[\frac{x e^{-x}}{-1} - e^{-x} \right]_0^{\infty} \times \left[\frac{e^{-y}}{-1} \right]_0^{\infty} = 1 \times 1 = 1$$

$$E(Y) = \int_0^{\infty} \int_0^{\infty} y f_{xy}(u, y) dx dy = 1 \quad V(X) = E(X^2) - (E(X))^2$$

$$\therefore \sigma_{xy} = E(XY) - E(X)E(Y) = 0 \text{ and } \rho_{xy} = \frac{\sigma_{xy}}{\sqrt{V(X)}\sqrt{V(Y)}} = 0$$

Verification:

$f_{xy}(x, y) = f_x(x) \cdot f_y(y)$, if x and y are independent

$$f_x(x) = \int_0^{\infty} f_{xy}(x, y) dy = e^{-x} \int_0^{\infty} e^{-y} dy = e^{-x} \left[\frac{e^{-y}}{-1} \right]_0^{\infty} = e^{-x}$$

$$f_y(y) = \int_0^{\infty} f_{xy}(x, y) dx = e^{-y} \left[\frac{e^{-x}}{-1} \right]_0^{\infty} = e^{-y}$$

$\therefore f_{xy}(x, y) = f_x(x) \cdot f_y(y) \Rightarrow x$ and y are independent.

(5.72)

$$\int_0^3 \int_0^x cxy dx dy = 1 \Rightarrow \int_0^3 cx \left[\frac{y^2}{2} \right]_0^x dx = 1 \Rightarrow \int_0^3 \frac{cx^3}{2} dx = 1$$

$$\Rightarrow \frac{c}{2} \left[\frac{x^4}{4} \right]_0^3 = 1 \Rightarrow \frac{c}{8} \times 81 = 1 \Rightarrow c = \frac{8}{81}$$

$$E(XY) = \frac{8}{81} \int_0^3 \int_0^x x^2 y^2 dy dx = \frac{8}{81} \int_0^3 x^2 \left[\frac{y^3}{3} \right]_0^x dx = \frac{8}{81} \left[\frac{x^5}{5} \right]_0^3 = \frac{8}{81} \times \frac{243}{5} = \frac{96}{25}$$

$$E(X) = \frac{8}{81} \int_0^3 \int_0^x x^2 y dy dx = \frac{8}{81} \int_0^3 x^2 \left[\frac{y^2}{2} \right]_0^x dx = \frac{4}{81} \left[\frac{x^5}{5} \right]_0^3 = \frac{4 \times 243}{81 \times 5} = \frac{12}{5}$$

$$E(Y) = \frac{8}{81} \int_0^3 \int_0^x xy^2 dy dx = \frac{8}{81} \int_0^3 x \left[\frac{y^3}{3} \right]_0^x dx = \frac{8}{81} \left[\frac{x^5}{5} \right]_0^3 = \frac{8 \times 243}{81 \times 5} = \frac{8}{5}$$

$$\therefore \sigma_{xy} = E(XY) - E(X)E(Y) = \frac{96}{25} - \frac{24}{25} = \frac{72}{25} = \frac{4}{25}$$

$$V(X) = E(X^2) - [E(X)]^2 = \frac{8}{81} \int_0^3 \int_0^x x^3 y dy dx = \frac{8}{81} \int_0^3 x^3 \left[\frac{y^2}{2} \right]_0^x dx = \frac{4}{81} \left[\frac{x^6}{6} \right]_0^3 = \frac{4}{81} \times \frac{729}{6} = 6$$

$$\therefore V(X) = 6 - \frac{144}{25} = \frac{150 - 144}{25} = \frac{6}{25}$$

$$V(Y) = E(Y^2) - [E(Y)]^2 = \frac{8}{81} \int_0^3 \int_0^x xy^3 dy dx = \frac{8}{81} \int_0^3 x \left[\frac{y^4}{4} \right]_0^x dx = \frac{2}{81} \left[\frac{x^6}{6} \right]_0^3 = 3$$

$$\therefore V(Y) = 3 - \frac{64}{25} = \frac{75 - 64}{25} = \frac{11}{25}$$

$$\therefore \rho_{xy} = \frac{\sigma_{xy}}{\sqrt{V(X)}\sqrt{V(Y)}} = \frac{4/25}{\sqrt{6/25} \sqrt{11/25}} = \frac{4}{\sqrt{66}} = 0.492368$$

We must have $\int_0^5 \int_{x-1}^{x+1} c \, dy \, dx = 1$

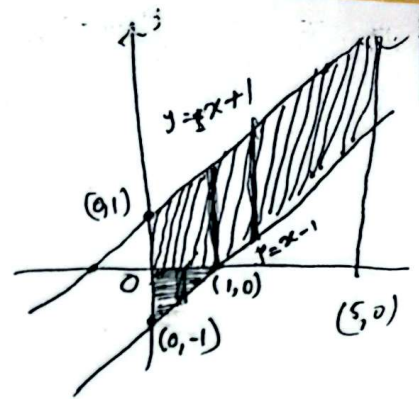
$$\Rightarrow \int_0^5 \int_{x-1}^{x+1} c \, dy \, dx - \int_0^1 \int_{x-1}^0 c \, dy \, dx = 1$$

$$\Rightarrow c \int_0^5 \left[y \right]_{x-1}^{x+1} dx - c \int_0^1 \left[y \right]_{x-1}^0 dx = 1$$

$$\Rightarrow c \int_0^5 2 \, dx - c \int_0^1 (1-x) \, dx = 1$$

$$\Rightarrow c \left[2x \right]_0^5 - c \left[x - \frac{x^2}{2} \right]_0^1 = 1$$

$$\Rightarrow c [10] - c \left[\frac{1}{2} \right] = 1 \Rightarrow \frac{19}{2} c = 1 \Rightarrow c = \frac{2}{19}$$



pdf of x and y is

$$E(xy) = \frac{2}{19} \left[\int_0^5 \int_{x-1}^{x+1} xy \, dy \, dx - \int_0^1 \int_{x-1}^0 xy \, dy \, dx \right]$$

$$= \frac{2}{19} \left[\int_0^5 x \left[\frac{y^2}{2} \right]_{x-1}^{x+1} dx - \int_0^1 x \left[\frac{y^2}{2} \right]_{x-1}^0 dx \right] = \frac{2}{19} \left[\int_0^5 \frac{4x^2}{2} dx - \int_0^1 \frac{-x^3 + 2x^2}{2} dx \right]$$

$$= \frac{2}{19} \left[\left[\frac{2x^3}{3} \right]_0^5 - \frac{1}{2} \left[\frac{-x^4}{4} + \frac{2x^3}{3} - \frac{x^2}{2} \right]_0^1 \right] = \frac{2}{19} \left[\frac{250}{3} - \frac{1}{2} \left(-\frac{1}{4} + \frac{2}{3} - \frac{1}{2} \right) \right]$$

$$= \frac{2}{19} \left[\frac{250}{3} + \frac{1}{24} \right] = \frac{2}{19} \left[\frac{2001}{24} \right] = \frac{2001}{19 \times 12}$$

$$E(x) = \frac{2}{19} \left[\int_0^5 \int_{x-1}^{x+1} x \, dy \, dx - \int_0^1 \int_{x-1}^0 x \, dy \, dx \right] = \frac{2}{19} \left[\int_0^5 2x \, dx - \int_0^1 (x-x^2) \, dx \right]$$

$$= \frac{2}{19} \left[25 - \frac{1}{6} \right] = \frac{2}{19} \times \frac{149}{6} = \frac{149}{57}$$

$$E(y) = \frac{2}{19} \left[\int_0^5 \left[\frac{y^2}{2} \right]_{x-1}^{x+1} dx - \int_0^1 \left[\frac{y^2}{2} \right]_{x-1}^0 dx \right] = \frac{2}{19} \left[\int_0^5 2x \, dx + \frac{1}{2} \int_0^1 (x^2 - 2x + 1) \, dx \right]$$

$$= \frac{2}{19} \left[x^2 \right]_0^5 + \frac{1}{2} \left[\frac{x^3}{3} - x^2 + x \right]_0^1 = \frac{2}{19} \left[25 + \frac{1}{2} \left(\frac{1}{3} - 1 + 1 \right) \right] = \frac{2}{19} \left[25 + \frac{1}{6} \right]$$

$$= \frac{2}{19} \times \frac{151}{6} = \frac{151}{57}$$

$$\therefore \sigma_{xy} = E(xy) - E(x)E(y) = \frac{2001}{19 \times 12} - \frac{149 \times 151}{57 \times 57} = \frac{104720}{108516}$$

$$E(x^2) = \frac{2}{19} \left[\int_0^5 \int_{x-1}^x x^2 dy dx - \int_0^1 \int_{x-1}^x x^2 dy dx \right] = \frac{2}{19} \left[\int_0^5 2x^2 dx - \int_0^1 (x^2 - x^2) dx \right]$$

$$= \frac{2}{19} \left\{ \left[\frac{2}{3} x^3 \right]_0^5 - \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 \right\} = \frac{2}{19} \left\{ \frac{250}{3} - \left(\frac{1}{3} - \frac{1}{4} \right) \right\}$$

$$= \frac{2}{19} \left\{ \frac{250}{3} - \frac{1}{12} \right\} = \frac{2}{19} \times \frac{1000-1}{12} = \frac{999}{114}$$

$$E(y^2) = \frac{2}{19} \left[\int_0^5 \int_{x-1}^{x+1} y^2 dy dx - \int_0^1 \int_{x-1}^0 y^2 dy dx \right] = \frac{2}{19} \left[\int_0^5 \left[\frac{y^3}{3} \right]_{x-1}^{x+1} dx - \int_0^1 \left[\frac{y^3}{3} \right]_{x-1}^0 dx \right]$$

$$= \frac{2}{19} \left[\frac{1}{3} \int_0^5 \{ (x+1)^3 - (x-1)^3 \} dx + \frac{1}{3} \int_0^1 (x-1)^3 dx \right]$$

$$= \frac{2}{57} \left[\int_0^5 (6x^2 + 2) dx + \int_0^1 (x^3 - 1 + 3x - 3x^2) dx \right]$$

$$= \frac{2}{57} \left\{ \left[\frac{6x^3}{3} + 2x \right]_0^5 + \left[\frac{x^4}{4} - x + \frac{3x^2}{2} - \frac{3x^3}{3} \right]_0^1 \right\}$$

$$= \frac{2}{57} \left\{ [250 + 10] + \left(\frac{1}{4} - 1 + \frac{3}{2} - 1 \right) \right\}$$

$$= \frac{2}{57} \left[260 + \frac{1}{4} \right] = \frac{2}{57} \times \frac{1040-1}{4} = \frac{1039}{114}$$

$$V(x) = E(x^2) - [E(x)]^2 = \frac{999}{114} - \frac{149^2}{57^2} = 1.929978$$

$$V(y) = E(y^2) - [E(y)]^2 = \frac{1039}{114} - \frac{151^2}{57^2} = 2.096183$$

$$\therefore \sigma_x = 1.389236 ; \sigma_y = 1.447820$$

$$\therefore \rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} = \frac{1.85416}{1.389236 \times 1.447820} = 0.920478$$